CONTRACT No. B-16317

INDEX

PROJECT	STRUCTURE	TYPE	SPAN	OVER	STATION
MAFR-158-2(5)	224-90-1545A	SUPERSTR. REPLACEMENT & WIDENING PRE STRESSED BOX BEAMS (SPREAD)	40'-0 SKEW 30° LT.	EIGHT MILE DITCH	STA. 703+55 LINE "A"

SHEET NO	SHEET DESIGNATION	SUBJECT	F.H.W.A. APPROVAL
1	ONE SHEET	INDEX AND TITLE SHEET	
2	ONE SHEET	TRAFFIC MAINTENANCE DETAILS	
3	R1	LAYOUT	
4	R2	GENERAL PLAN	
5	R3	ABUTMENT NO.1 & NO.2 DETAILS	
6	R4	ABUTMENT NO.2 DETAILS	
7	R5	ABUTMENT NO.1 & NO.2 DETAILS	
8	R6	BEAM DETAILS	
9	R7	SUPERSTRUCTURE DETAILS	
10	R8	SUPERSTRUCTURE DETAILS	
11	R9	APPROACH DETAILS	
12	ONE SHEET	BRIDGE SUMMARY	
13	ONE SHEET	ESTIMATE OF QUANTITIES	
14-17	FOUR SHEETS	CROSS-SECTIONS (LINE "A")	
10A	R8A	CONCRETE BARRIER RAILING DETAILS	
10B	R8B	CONCRETE BARRIER RAILING DETAILS	

INDIANA DEPARTMENT OF HIGHWAYS

BRIDGE PLANS

FOR SPANS OVER 20 FEET ON U.S. 224 WELLS COUNTY PROJECT MAFR-158-2(5) REPAIRS & WIDENING TO EXISTING BRIDGE

THIS STRUCTURE IS LOCATED ON U.S. 224 OVER EIGHT MILE DITCH IN SECTION 5, ON TOWNSHIP LINE T.27N. & T.28N., R.13 E., IN WELLS COUNTY, INDIANA.

TRAFFIC DATA		
A.D.T. (1983)		3500 V.P.D.
A.D.T. (19 PROJECTED)		V.P.D.
D.H.V. (19 PROJECTED)		V.P.D.
TRUCKS		D.H.V. % A.D.T. %
DESIGN SPEED		M.P.H.
ACCESS CONTROL		

PLANS PREPARED BY:

AMERICAN CONSULTING ENGINEERS INC.

INDIANAPOLIS, INDIANA

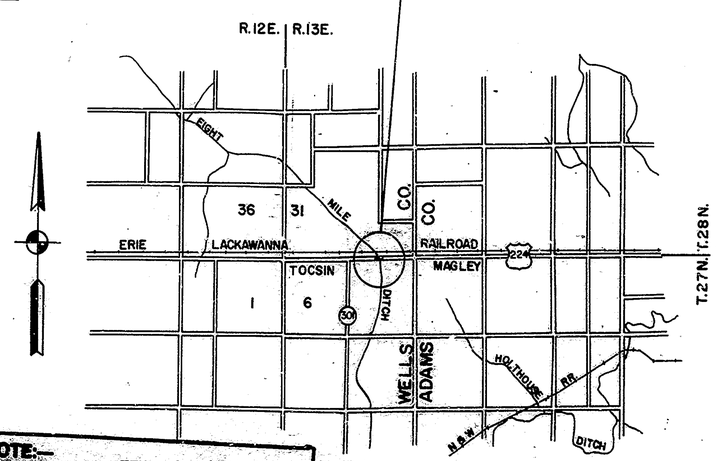
CERTIFIED: John S. Spencer

DATE: 1-25-85

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

NOTE:
WHEREVER "INDIANA STATE HIGHWAY COMMISSION" APPEARS IN THESE PLANS, IT SHALL BE INTERPRETED AS "INDIANA DEPARTMENT OF HIGHWAYS"

BRIDGE FILE: 224-90-1545A



LOCATION PLAN
Scale: 1 Inch = 1 Mile

NOTE:—
WHENEVER FR-158-2(5) APPEARS IN THESE PLANS OR CONTRACT DOCUMENTS IT SHALL BE INTERPRETED AS MAFR-158-2(5)

INDIANA DEPARTMENT OF HIGHWAYS STANDARD SPECIFICATIONS DATED 1985 TO BE USED WITH THESE PLANS.

REVISIONS	
DATE	SHEET NO.
4-17-85	1, 13
6-10-85	1, 3, 4, 12, 13, 15
7-16-85	1, 3, 10, 12, 14
8-6-85	1, 3
12-16-85	1, 3, 11, 12, 15
3-14-86	1, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 3, Added 10A, 10B, 21A, 34A, 34B
	Deleted: 18, 19, 20, 21, 34, 34, 34

REVISIONS	
DATE	SHEET NO.
6-10-87	1, 3, 4, 10, 12, 13, 15; Added 34B

BRIDGES OVER 20' SPAN

FEDERAL REGION NO.	STATE	PROJECT NO.	FISCAL YEAR	SHEET NO.	TOTAL SHEETS
5	IND.		198	1	43

INDEX CONTINUED
STANDARD DRAWINGS

SHEET NO.	SHEET DESIGNATION	SUBJECT	F.H.W.A. APPROVAL	ADOPTED Y-M-D OR REVISION
~~18~~	~~BRIDGE STD. BR1~~	~~ALUMINUM BRIDGE RAILING~~	PENDING	6-10-87
~~19~~	~~BRIDGE STD. BR2~~	~~ALUMINUM BRIDGE RAILING DETAILS~~	PENDING	6-10-87
~~20~~	~~BRIDGE STD. BR3~~	~~STEEL BRIDGE RAILING~~	10-16-80	R-11-3-80
~~21~~	~~BRIDGE STD. BR4~~	~~STEEL BRIDGE RAILING DETAILS~~	2-10-78	R-12-1-78
21A	BRIDGE STD. BR5	RAILING CONNECTION DETAILS		
	BRIDGE STD. BR6	RAILING CONNECTION DETAILS		
22	BRIDGE STD. C1	MISCELLANEOUS DETAILS	12-21-81	R 12-7-81
	BRIDGE STD. C2	MISCELLANEOUS DETAILS		
23	BRIDGE STD. C3	MISCELLANEOUS DETAILS	12-21-81	R 12-7-81
	BRIDGE STD. C4	MISCELLANEOUS DETAILS		
	BRIDGE STD. D	CASTING DETAILS ROADWAY DRAINS		
	BRIDGE STD.			
	BRIDGE STD.			
	BRIDGE STD. PB	PRESTRESSED CONCRETE TYPE I-BEAMS		
	BRIDGE STD. PB	PRESTRESSED CONCRETE TYPE I-BEAMS		
24	BRIDGE STD. PB6	PRESTRESSED BOX BEAMS	7-26-71	R 3-1-71
	BRIDGE STD. PB	PRESTRESSED COMPOSITE BOX BEAMS WIDE		
25	BRIDGE STD. PB9B	PRESTRESSED COMPOSITE BOX BEAMS 4' WIDE	7-26-71	R 3-1-71
26	BRIDGE STD. PB10	TOLERANCES FOR FABRICATION OF PRESTRESSED BEAMS	8-14-63	A NOV. 1962
27	BRIDGE STD. PB11	ELASTOMERIC BEARING PAD DETAILS	10-19-83	R 9-9-83
	BRIDGE STD.			
	BRIDGE STD.			
	BRIDGE STD. R2A	BRIDGE LIGHTING DETAILS		
	BRIDGE STD. R2B			
	BRIDGE STD. S1	MISCELLANEOUS DETAILS		
	BRIDGE STD. SH1	STEEL SHOE DETAILS		
	BRIDGE STD. T SHEET A	STANDARD TEMPORARY BRIDGE		
	BRIDGE STD. T SHEET B	STANDARD TEMPORARY BRIDGE		
	BRIDGE STD.			
	BRIDGE STD.			
	BRIDGE STD.			
	BRIDGE STD.			
	ROAD STD. SHEET A	STANDARD PAVEMENT JOINTS		
	ROAD STD. SHEET B	STANDARD PAVEMENT JOINTS		
28	ROAD STD. SHEET MA	MISCELLANEOUS STANDARDS		R-9-4-84
	ROAD STD. SHEET MA	MISCELLANEOUS STANDARDS		
29	ROAD STD. SHEET MB	MISCELLANEOUS STANDARDS		R-9-4-84
	ROAD STD. SHEET MB2	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MC	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MC1	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MD	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MD	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MD	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET ME	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET ME	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MH	MISCELLANEOUS STANDARDS		
30	ROAD STD. SHEET MH1	MISCELLANEOUS STANDARDS	PENDING	R-6-3-85
	ROAD STD. SHEET MI	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MN	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
31	ROAD STD. SHEET MP	MISCELLANEOUS STANDARDS		R-6-3-85
	ROAD STD. SHEET	MISCELLANEOUS STANDARDS		
	ROAD STD. SHEET MQ	MISCELLANEOUS STANDARDS		
32	ROAD STD. SHEET MT7	MISCELLANEOUS STANDARDS	7-19-85	R 6-3-85
33	ROAD STD. SHEET MT11	MISCELLANEOUS STANDARDS	12-27-82	A NOV. 1982
	ROAD STD.	STANDARD REINF. CONCRETE BOX CULVERTS		
	ROAD STD.	STANDARD REINF. CONCRETE CULVERTS		
~~34~~	~~ROAD STD. SHEET GR4~~	~~GUARD RAIL CLASS GA~~	PENDING	R-2-2-84
34A	ROAD STD. SHEET GR4A	GUARD RAIL CLASS HS		R-2-2-84
34B	ROAD STD. SHEET GR10B	GUARD RAIL CLASS W-BEAM GUARD RAIL TERMINAL SECTION	PENDING	A-12-01-86
~~35~~	~~ROAD STD. SHEET GR5~~	~~GUARD RAIL PIER DETAILS~~	5-21-82	R-4-1-83
~~36~~	~~ROAD STD. SHEET GR6~~	~~STEEL TYPE GUARD RAIL DETAILS~~		
35A	ROAD STD. SHEET GR7	GUARD RAIL PIER CONNECTION DETAIL	5-21-82	R-4-1-83
37	ROAD STD. SHEET GR10	GUARD RAIL ANCHOR ENDS		R-2-1-85
38	ROAD STD. SHEET CB-2	TEMPORARY CONCRETE BARRIER	PENDING	R6-1-84
39	ROAD STD. SHEET MT3	TRAFFIC SIGN DETAILS	8-30-82	A JULY 1982
	ROAD STD. SHEET 1 DETOURS	STANDARD DETOUR SIGNS		
	ROAD STD. SHEET 1A DETOURS	STANDARD DETOUR SIGNS		
	ROAD STD. SHEET 1B DETOURS	STANDARD DETOUR SIGNS		
	ROAD STD. SHEET 2 DETOURS	STANDARD DETOUR SIGNS		
40	ROAD STD. SHEET 2A DETOURS	STANDARD DETOUR SIGNS	4-1-85	R-2-1-85
41	ROAD STD. SHEET 3 DETOURS	STANDARD DETOUR SIGNS	4-10-84	R-2-1-84
	ROAD STD. SHEET 3A DETOURS	STANDARD DETOUR SIGNS		
42	ROAD STD. SHEET 4 DETOURS	STANDARD DETOUR SIGNS	12-8-83	R-10-3-83
43	ROAD STD. SHEET 5 DETOURS	STANDARD DETOUR SIGNS	4-10-84	R-2-1-84
	ROAD STD. SHEET 5A DETOURS	STANDARD DETOUR SIGNS		

APPROVED 3-13-85

CHIEF HIGHWAY ENGINEER — INDIANA

RECOMMENDED FOR APPROVAL 3-11-85

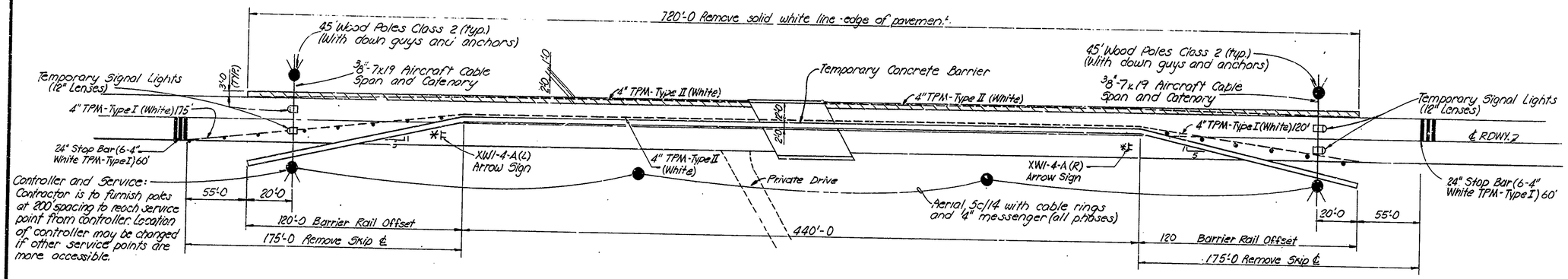
WILLIAM B. ABBOTT REGISTERED No. 8643 STATE OF INDIANA PROFESSIONAL ENGINEER

FEDERAL HIGHWAY ADMINISTRATION
DEPARTMENT OF TRANSPORTATION

APPROVED:

DIVISION ADMINISTRATOR — DATE

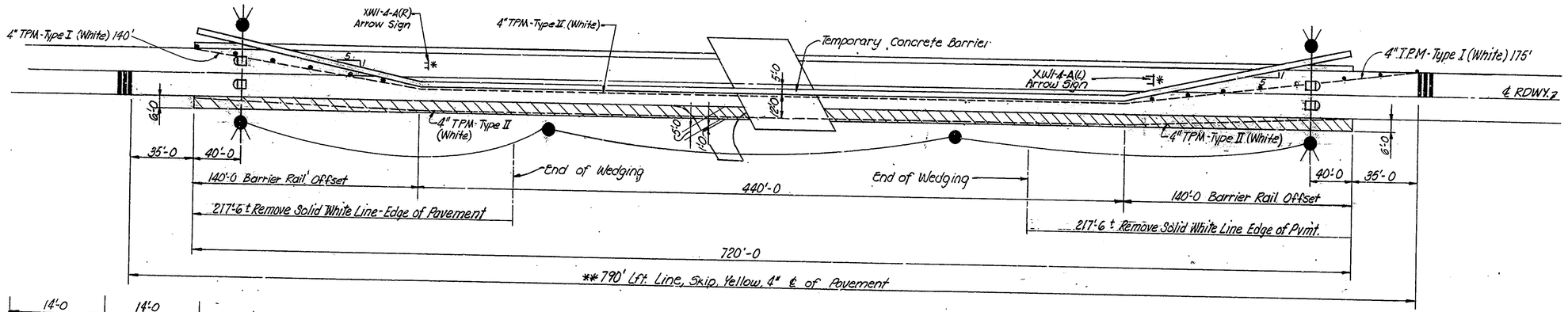
BRIDGE FILE: 224-90-1545A

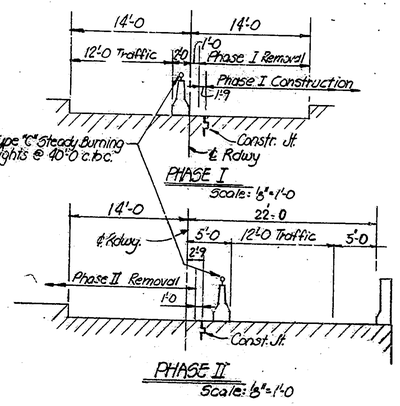


NOTE:
Wood Poles to be Placed as Far as Possible From the edge of the Pavement.

PHASE I

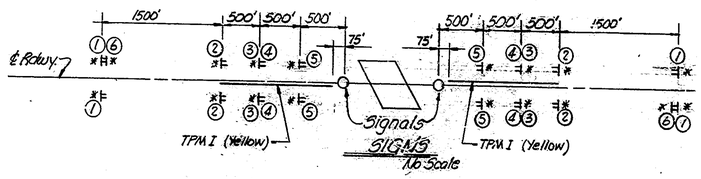
** To be placed after Phase II construction.





PHASE II

** To be placed after Phase II construction.



NOTES:

Type "C" Steady Burning Lights shall be mounted on top of temporary portable concrete barrier where barrier is used to define an edge of lane. Spacing of lights shall be numerically equal to the posted speed with a maximum spacing of 40 feet.

Approved lights shall mark barricades and signs continuously at night.

All signs shall be mounted in such a manner that the vertical distance between the bottom of the sign and the edge of the pavement shall not be less than 5 feet and horizontal distance from the edge of pavement to the edge of sign shall not be less than 12 feet (R/W permitting) or 6 feet from edge of a paved shoulder.

SIGN LEGEND

① Road Construction Ahead W20-1
② Do Not Pass R4-1-A
③ One Lane Road Ahead W20-4
④ Advisory Speed Limit XW13-1-A
⑤ Signal Ahead XW3-3a-A
⑥ End Construction G20-2
* Low Intensity Flashing Yellow Light Type A.

SIGNALIZATION NOTES

All signal equipment to remain the property of the contractor. Cost of equipment, materials and labor (including poles, cables, signal equipment, and hardware) needed to install, operate and remove traffic signal system, shall be included in the pay item "Temporary Traffic Signals." (L. Sum.)
Aerial cable may be placed on either the east or west side of structure at the option of the contractor.

TPM-TYPE I

WHITE: 730 LFT.
YELLOW: 2850 LFT.

Paint edges of pavement within construction limits with solid white line.

LEGEND

Bituminous Widening (990#/Syd Base Size 50.) To be paid for as "Bituminous Mixture for Approaches."

° Drum or Type II Barricades w/ Type "C" Steady burning light.

TPM Temporary Pavement Marking, Type and Color as Noted.

DESIGNED ____ C'K'D. ____
DRAWN Ranger C'K'D. WJW 4-27-84
TRACED ____ C'K'D. ____

Rev: 3-A-86, Metal or Type I Barricades deleted.

TRAFFIC MAINTENANCE DETAILS
INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted DATE:- JANUARY 25, 1985
SUBMITTED FOR APPROVAL John S. Spencer
DRAWING:- OF SHEET:- 2 OF 43
PROJECT:- FR-158-2(5)
CONTRACT NO. B-16317
BRIDGE FILE:- 224-90-1545A

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

SEC. 32, T28N, R13E
JEFFERSON TWP.
WELLS CO.

MICHAEL L. & KATHLEEN L. CHAMBERLAIN

ROBERT E. & OLIVE JOHNLOZ

SEC. 5, T27N, R13E
LANCASTER TWP.
WELLS CO.

BEGIN PROJECT Nº FR-158-2( )
Sta. 701+70.0 LINE "A"

END PROJECT Nº FR-158-2( )
Sta. 705+85.00 LINE "A"

J. DOYLE & DOROTHY M. BARGER

LINE "A"
BRG. N. 88° 35' E.

TWP. & SECTION LINE (BRG. N88°35'E)

U.S. # 224 (Blk. Top) 90'

Steel G.R.

STRUCTURE Nº 10
30-18'0 PIPE
38

STRUCTURE Nº 11
185' 18'0 PIPE

Construction Limits

Exist. Paved Side Ditch

Remove and Replace P.S.D. with Paved Side Ditch

Skew 30° LT.

℄ Structure Sta. 703+55

Special Class II Drive Sta. 703+13 Rt. W=12'

Special Class II Drive Sta. 703+26 Rt. W=12'

Special Class II Drive Sta. 703+10 Rt.

TAPER SHLDR. TO MATCH EXIST. (TYP.)

REMOVE EXIST. PIPE

REMOVE EXIST. FIELD TILE

14" F. Tile

6" Gas Line

Start Fence

Priv. Dr. (C. Stn.)

Field Ent. (Dirt)

(Soybeans)

(Lawn)

(Corn)

(REMOVE EXIST. PIPE)

TEMP R/W FOR DRIVE CONSTR.

TEMP R/W FOR PIPE REMOVAL

105' TEMP. R/W FOR PIPE REMOVAL

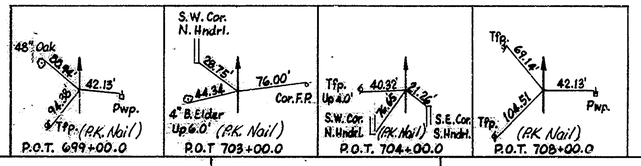
12" F. Tile Broken 5' into Bank

Note: Removal of exist. pipes to be included in the cost of other items.

All R/W on this sheet to be as shown.
All R/W described from Line "A".

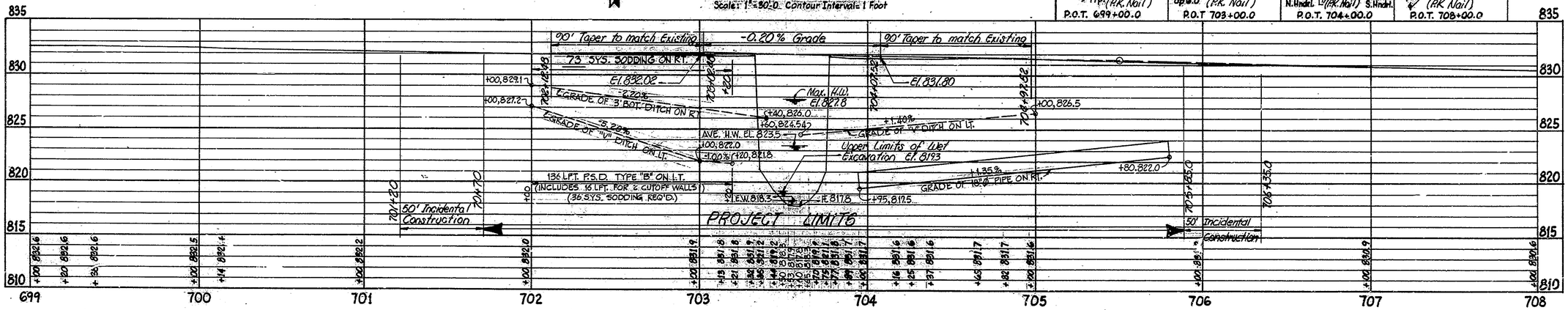
PRESENT STRUCTURE

The existing structure is an R.C. Girder Bridge with one span at 40'0, skew 30° Lt. with a 28'0 clear roadway. The structure was built originally in 1937.



SITUATION PLAN

Scale: 1"=30'-0 Contour Interval = 1 Foot



BENCH MARK DATA

BM #1 Br. El. 831.91, Bt. sp. in Pwp. 96' Lt. of Sta. 700+80 "A"
BM #2 Br. El. 831.41, □ cut in Top S.E. Wing and Abut., 16' Rt. Sta. 703+84 "A"
BM #3 Br. El. 826.98, Bt. sp. in Tfp., 40' Lt. of Sta. 707+04 "A"
USC & GS BM-J-35 (1934) El. 834.94, Br. Disc. in NW Quadrant of Erie RR and Co. Rd.

NOTE:
Project 349 & Br. Survey datum is ± 0.06' above USC & GS datum.

PROFILE

Scales: Horiz. 1"=30'-0, Vert. 1"=5'-0

Note: 15 LFT. OF 15" ϕ PIPE @ STA. 704+06 AND 15 LFT. OF 12" ϕ PIPE @ STA. 703+51 TO EXTEND EXISTING FIELD TILE (UNDISTRIBUTED QUANTITY)

EARTHWORK

Fill + 20%	= 680	Cys.
- Common Excavation	= 60	Cys.
- Surplus Excavation	= 130	Cys.
Borrow	490	Cys

ϕ Benching (Estimated) 280 Cys

ϕ No direct payment. Cost of Benching to be included in the cost of other items in the Contract.

UTILITY OWNERS

POWER
United REMC
State Road 5 South
Huntington, IN. 46750

TELEPHONE
United Telephone Co. of Indiana
248 W. Monroe
Decatur Indiana

GAS
NIPSCO
Ft. Wayne Indiana

Field Notes: See Book BR-2418, pp. 1-19.

Rev. 1-14-87 RELOCATE DRIVE AT SW COR. OF STRUCTURE. REVISE R/W.
Rev. 4-25-86 Delete I & M Electric Co. and change to United REMC.
Rev. 4-17-85 Add 12" ϕ PIPE, 18" ϕ PIPE
Rev. 6-17-85 Revise drive @ Sta. 703+26 Rt; Revise Str. No. 10
Rev. 7-11-85 Delete Class II Drives @ Stas. 705+30 Lt. & 705+55 Rt., Delete Temp. R/W For Drive Constr on North Side of Road
Rev. 8-6-85 Revise R/W @ Class II Drive @ Sta. 703+26 Rt
Rev. 12-16-85 Delete Spl. Class II Drive Sta. 703+26 Rt., Add Spl. Class II Drive Sta. 703+10 Rt.

LAYOUT

SUPERSTRUCTURE REPLACEMENT & WIDENING

1 SPAN @ 40'-0 SKEW 30° LT. — 44'-0 ROADWAY
OVER EIGHT MILE DITCH — ON U.S. 224

INDIANA STATE HIGHWAY COMMISSION

WELLS COUNTY

SCALE:- As Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL John S. Spencer

DRAWING:- R1 OF R9 SHEET:- 3 OF 43

PROJECT:- FR-158-2(5)

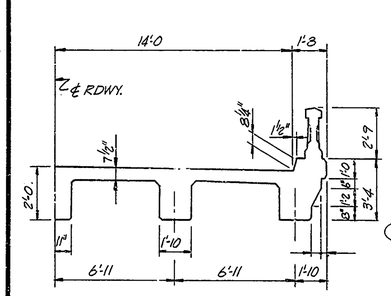
CONTRACT NO. B-16317

BRIDGE FILE:- 224-90-1545A

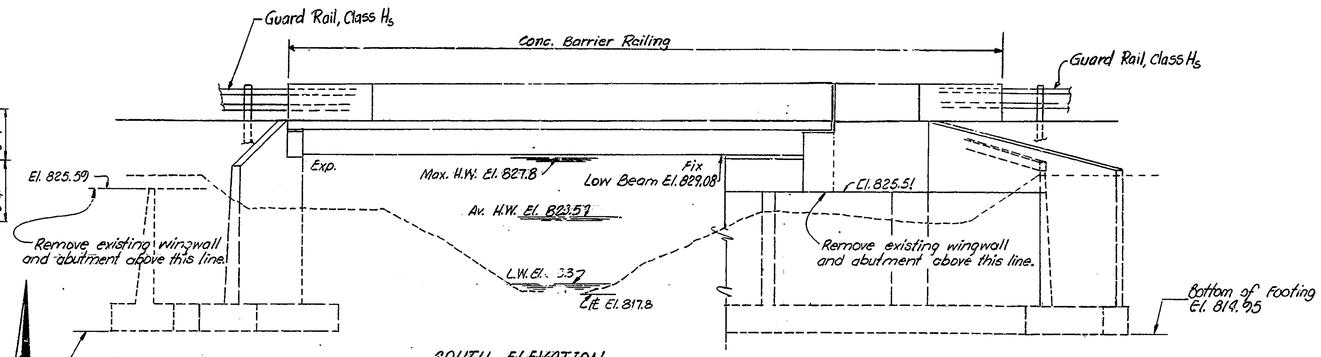
JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

DESIGNED ____ C'K'D ____
DRAWN B.G.E. C'K'D MJW 4-27-84
TRACED ____ C'K'D ____

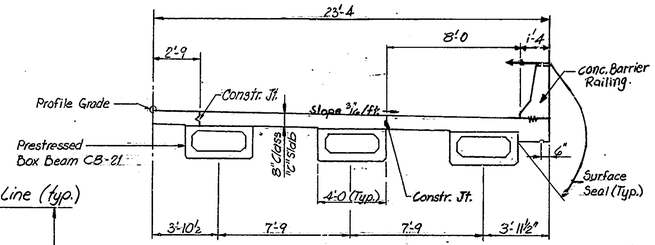
STRUCTURE BUILT TO A -0.200% GRADE



HALF SECTION (EXISTING STRUCTURE)
Scale: ¼"=1'-0

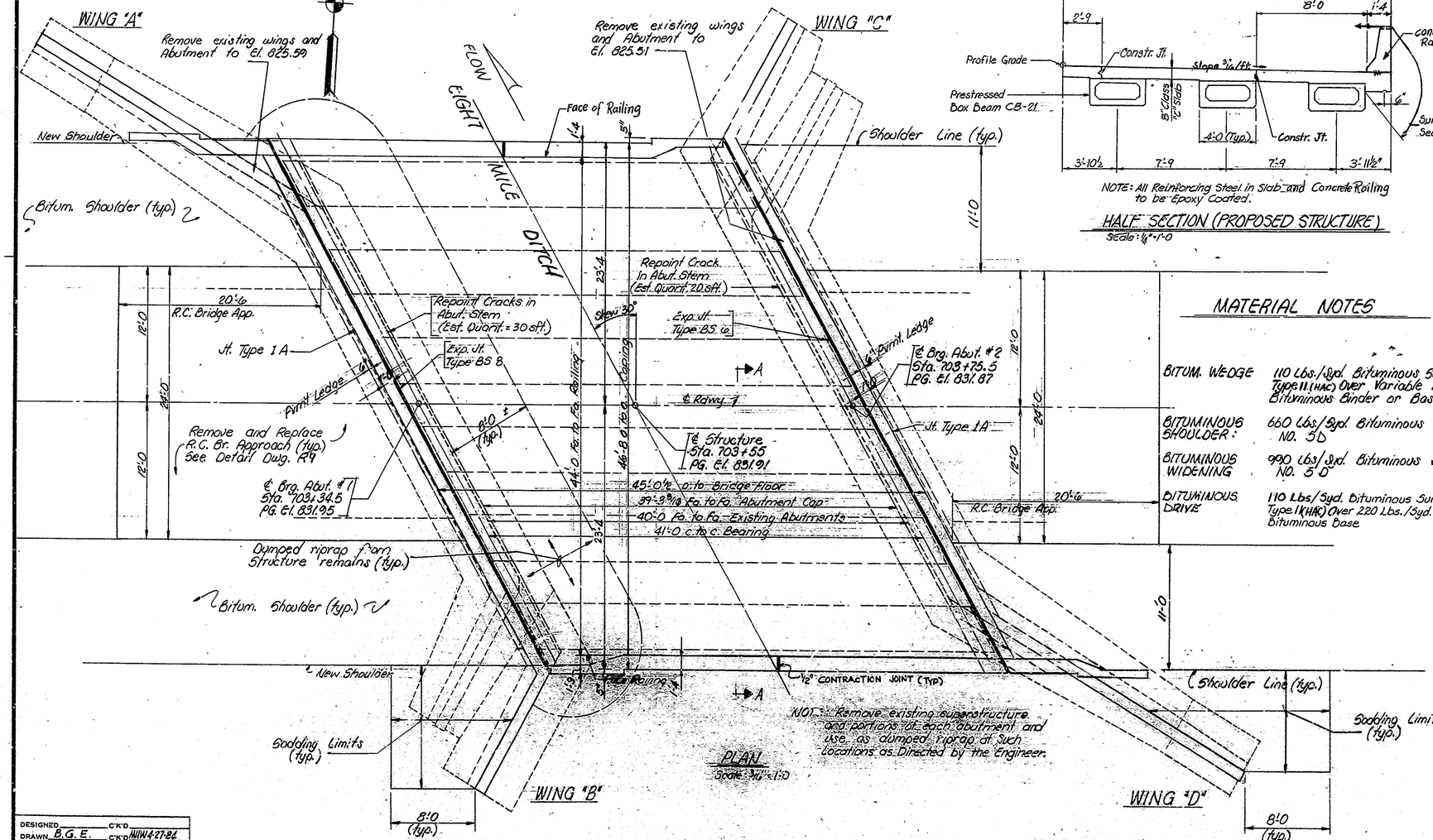


SOUTH ELEVATION
Scale: 3/16"=1'-0



NOTE: All Reinforcing Steel in Slab and Concrete Railing to be Epoxy Coated.

HALF SECTION (PROPOSED STRUCTURE)
Scale: ¼"=1'-0



MATERIAL NOTES

BITUM. WEDGE	110 Lbs./Syd. Bituminous Surface Type II (HAC) Over Variable Depth Bituminous Binder or Base.
BITUMINOUS SHOULDER:	660 Lbs./Syd. Bituminous Base NO. 5D
BITUMINOUS WIDENING	990 Lbs./Syd. Bituminous Base NO. 5D
BITUMINOUS DRIVE	110 Lbs./Syd. Bituminous Surface Type I (HAC) Over 220 Lbs./Syd. Bituminous Base

GENERAL NOTES

Depth of footings to be extended if found necessary. See Article 206.11 (c) of the Specifications.

Plans for the Existing Structure (224-90-1545) are on file in the Central Office and are available upon request.

Seal all Joints and Cracks in the Approach Pavement within the Limits of the Bituminous Wedges With a hot poured Joint Sealer before placing Bituminous Wedge.

Reinforcing steel covering shall be 2½ inches in top and 1 inch minimum in bottom of floor slabs, 3 inches in footings except bottom steel which shall be 4 inches and 2 inches in all other parts unless noted.

Concrete in footing to be Class "B"

Concrete in Superstructure and Conc. Barrier Rail to be class "C."

Concrete in abutment stems, and caps and wingwalls to be Class "A."

Guard Rail removed on this Project to be disposed of by Contractor.

Continuous concrete pours shall be required between construction joints as shown on detail plans.

Waterproof backs of abutments and wingwalls in accordance with Article 702.22 of the Specifications.

Chamfer exposed edges 1" unless noted.

Surface Seal the top of Abutment Caps, top of Deck, face of Deck coping, underside of bridge floor from coping to face of outside beam, top and front face of mudwalls, Conc. Barrier Rail and 1'-0 of R.C. Bridge Approach.

Where new work is to be fitted to old work, the Contractor shall check all dimensions and conditions in the field and report any errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of new part to old.

All bituminous material in this contract to be included in the pay item, "Bituminous Mixture for Approaches," except tack coat which is paid for separately in square yards.

DESIGN DATA

Designed for HS 20-44 loading in accordance with 1983 A.A.S.H.T.O. Specifications and Subsequent Interim Specifications.

STANDARD DRAWINGS

Br. Std.	Rd. Std.	Description
BR6		Concrete Barrier Railings
C1		Reinforcing Bars, Notes, 4"Ø Drains
CB		Construction Joints, Type A & Type IA Joints
PB6		Prestressed Box Beams
PB7B		Prestressed Composite Box Beams
PB10		Tolerances For Fabrication of Prestressed Beams
PB11		Elastomeric Bearing Pad Details
	MA	Reinforced Concrete Bridge Approach
	MB	Paved Side Ditch
	MH1	Class II Drive, Class V Field Entrance
	GR4A	Guard Rail Class Hs
	GR7	Bracket Detail
	GR10	G.R.E.T. Type I
	CB2	Temporary Concrete Barrier
	GR10B	W-Beam Guard Rail Terminal Section
	Sht. 1 of 5 Det	Standard Detour Sheets
	Sht. 4 Det	Standard Detour Sheet
	Sht. 5 Det	Standard Detour Sheet
	MP	Group "D" Pipe
	MT-7	Controller mounted on wood pole
	MT-11	Span & Catenary Suspension
	MT-3	Traffic Sign Details

SUPERSTRUCTURE REPLACEMENT & WIDENING

GENERAL PLAN

1 SPAN @ 41'-0 SKEW 30° LT. 44'-0 ROADWAY
OVER EIGHT MILE DITCH ON S.R. 224

INDIANA STATE HIGHWAY COMMISSION

WELLS COUNTY

SCALE:- As Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL John S. Spencer

DRAWING:- R2 OF R9 SHEET:- 4 OF 43

PROJECT:- FR-158-2(5)

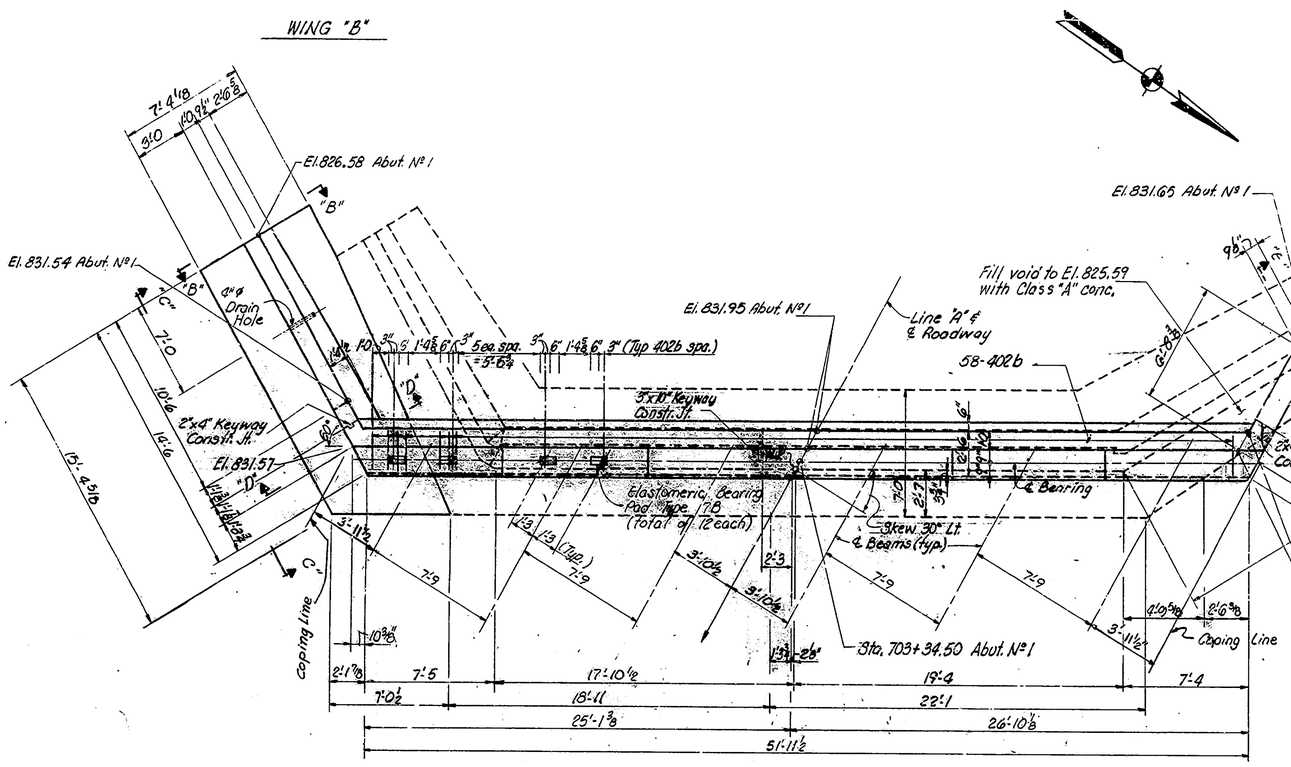
CONTRACT NO. B-16317

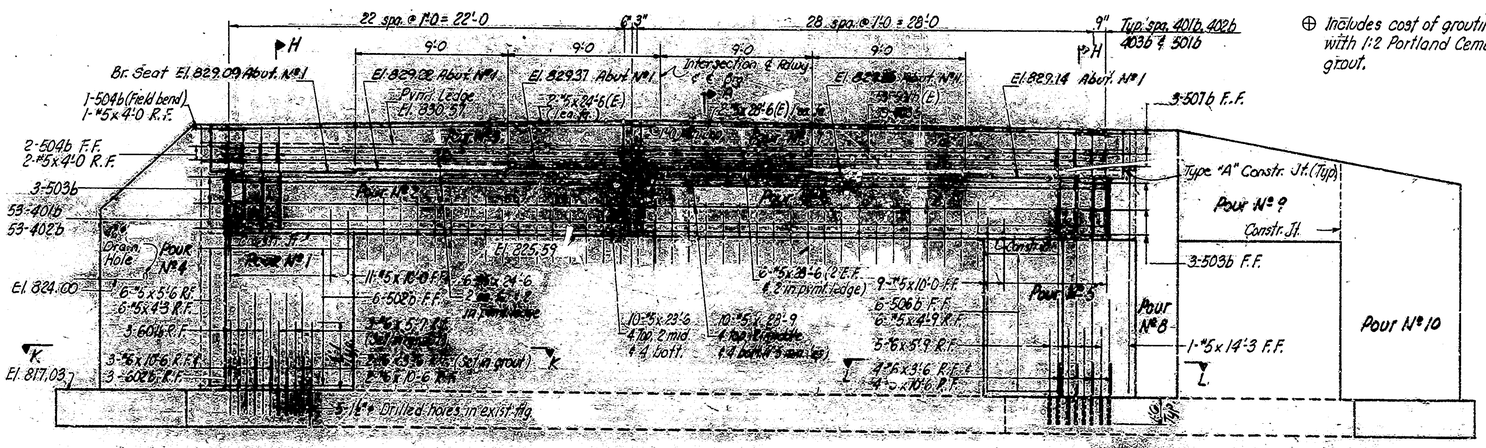
BRIDGE FILE:- 224-90-1545A

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

DESIGNED ____ C'K'D ____
DRAWN B.G.E. C'K'D WHW 4-27-84
TRACED ____ C'K'D ____

Material Notes, Out to Out Distance.
Rev: 3-4-86, Conc. Barrier Railing, General Notes, standard Drawings.
Rev: 01-14-87; Added Rd. Std. GR10B





⊕ Includes cost of grouting with 1:2 Portland Cement grout.

Notes:
See Bridge Standard C.1 for Reinforcing Bar Notes.
See Bridge Standard C.3 for Type "A" Constr. Jt.
See Dwg. R4 & R5 for additional details.
See Bridge Standard PB11 for Elastomeric Bearing Pad Details.
E Indicates epoxy coated reinf. steel.
* Indicates Abutment N° 1 only.
** Indicates Abutment N° 2 only.

Rev: 2-25-86 Coping line

BILL OF MATERIALS
ABUTMENT N° 1

ABUT. N° 2 SAME UNLESS NOTED

Mark or Size	No of Bars	Length (Ft.)	Weight (Lbs.)
Reinforcing Steel			
Epoxy Coated Reinf. Steel			
501b (E)	53	10'-1	
#5 (E)	2	28'-6	
#5 (E)	2	29'-6	
Total Epoxy Coated Reinf. Steel			668
Plain Reinf. Steel			
701b	16	5'-4	
702b	12	8'-0	
703b	4	23'-8	
704b	3	22'-0	
705b	12	12'-7	
#7	2	14'-3	
#7	10	6'-9	
#7	16	5'-0	
#7	2	4'-3	
#7	2	4'-0	
#7	2	2'-6	
	Total #7		1411
601b	3	6'-7	
602b	3	4'-4	
#6	9	10'-6	
#6	8	5'-9	
#6	6	3'-6	
	Total #6		292
502b	6	11'-0	
503b	6	5'-9	
504b	3	6'-0	
505b	4	23'-8	
506b	6	14'-0	
507b	5	7'-6	
508b	5	6'-9	
509b	3	22'-0	
510b	11	8'-11	
* #5	10	28'-9	
* #5	6	28'-6	
* #5	6	29'-6	
** #5	10	23'-6	
#5	2	18'-3	
#5	2	17'-6	
#5	2	15'-0	
#5	5	14'-3	
#5	6	12'-9	
#5	16	11'-3	
#5	20	10'-0	
#5	4	8'-9	
#5	24	7'-6	
#5	8	5'-6	
#5	2	5'-3	
#5	6	5'-0	
#5	6	4'-9	
#5	6	4'-3	
#5	11	4'-0	
	Total #5 Abut. N°1		2086
** #6	6	28'-6	
** #6	10	27'-9	
** #5	16	24'-6	
	Total #5 Abut N°2		2086
401b	53	9'-10	
402b	111	3'-6	
403b	53	5'-6	
	Total #4		731
Total Plain Reinf. Steel Abut. N°1			4520
Total Plain Reinf. Steel Abut. N° 2			4520

Concrete		
Class "A" Abut. N° 1		Quant.
Pour N° 1		6.1 Cys
Pour N° 2		8.4 "
Pour N° 3		2.4 "
Pour N° 4		8.1 "
Pour N° 5		5.3 "
Pour N° 6		10.6 "
Pour N° 7		3.1 "
Pour N° 8		2.9 "
Pour N° 9		2.0 Cys.
Pour N° 10		4.4 "
Pour in Void		1.9 "
Total Class "A" Abut N°1		55.2 Cys.
Total Class "A" in Footing		12.1 Cys.
Class "A" Abut. N° 2		
Pour N° 1		5.3 Cys.
Pour N° 2		8.7 "
Pour N° 3		2.5 "
Pour N° 4		2.9 "
Pour N° 5		2.0 "
Pour N° 6		4.4 "
Pour N° 7		6.1 "
Pour N° 8		10.3 "
Pour N° 9		3.0 "
Pour N° 10		8.1 "
" in Void		1.9 "
Total Class "A" in Abut. N° 2		55.2 Cys.
Miscellaneous		
⊕ Field Drilled Holes in Concrete	18 Ea.	
Surface Seal	709 Sft.	

ABUTMENT N° 1 & N° 2 DETAILS
INDIANA STATE HIGHWAY COMMISSION

SCALE:- 4" = 1'-0 Unless Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL John S. Spencer

DRAWING:- R3 OF R9 SHEET:- 5 OF 43

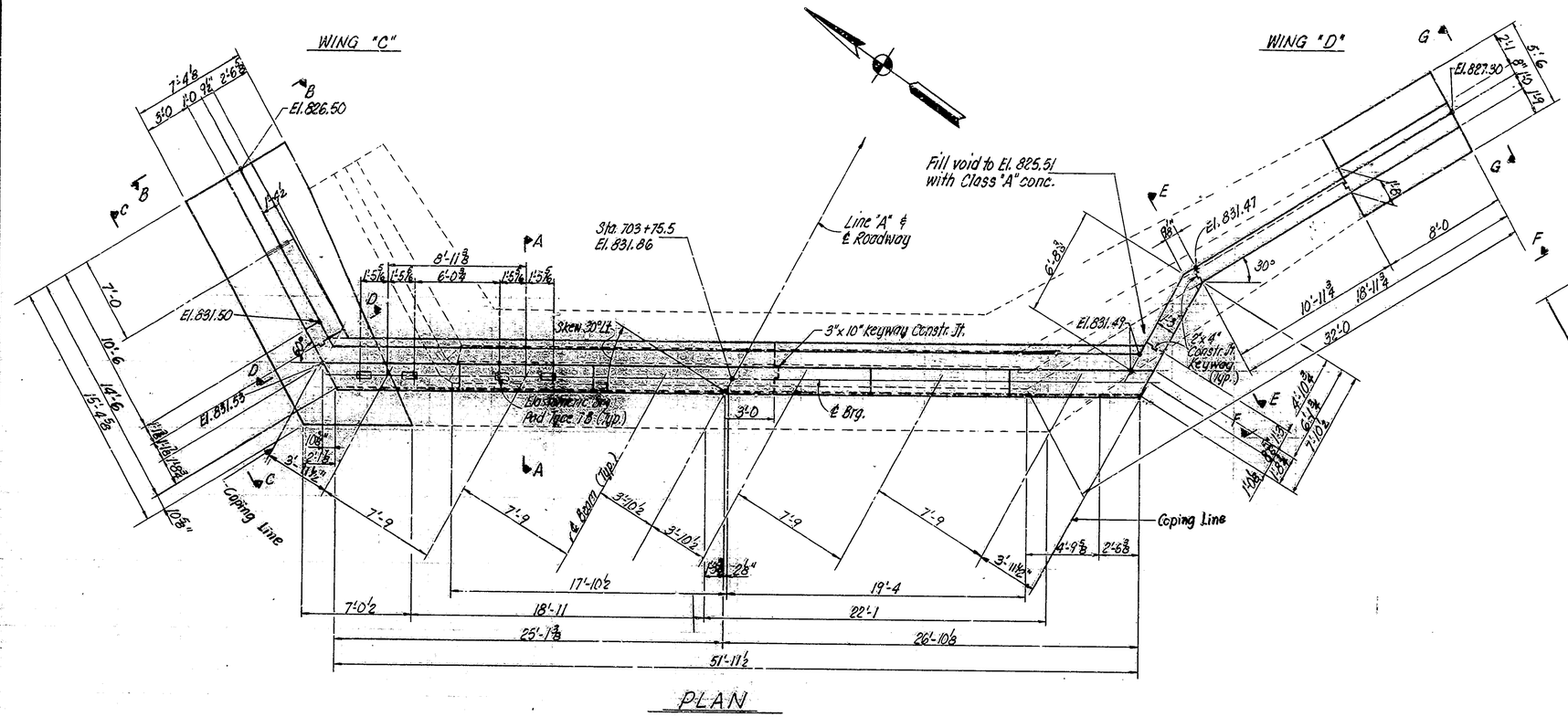
PROJECT:- FR-153-2(5)

CONTRACT NO. B-16317

BRIDGE FILE:- 224-90-1454

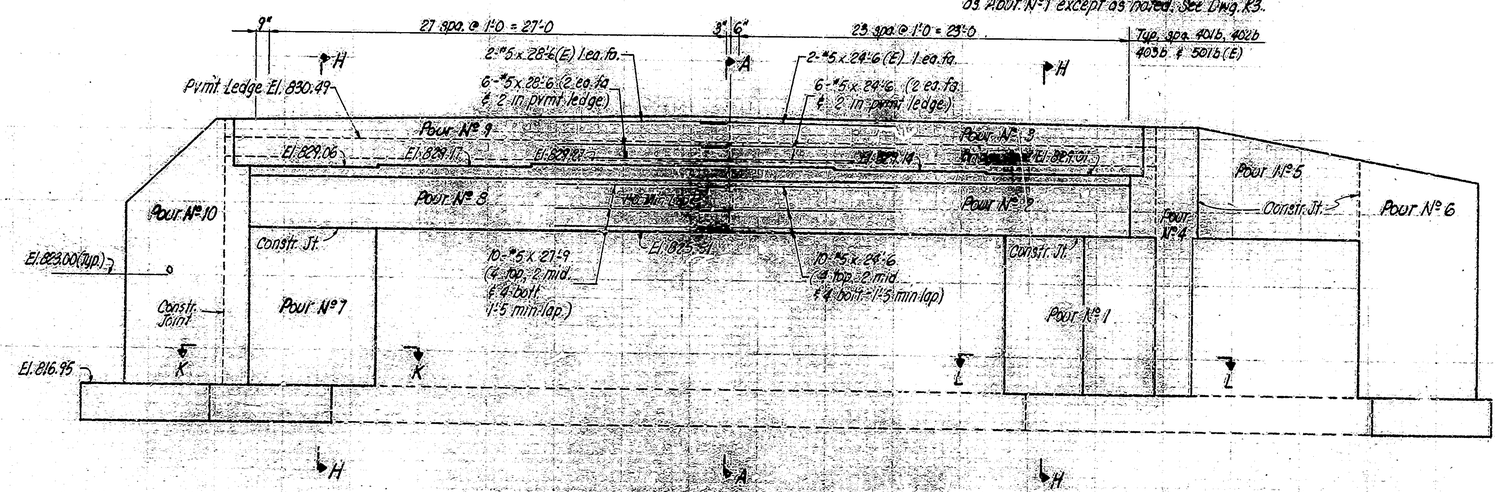
JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

DESIGNED		C'K'D	
DRAWN	M.E.B. 8-2-84	C'K'D	J.K. 9-10-84
TRACED		C'K'D	



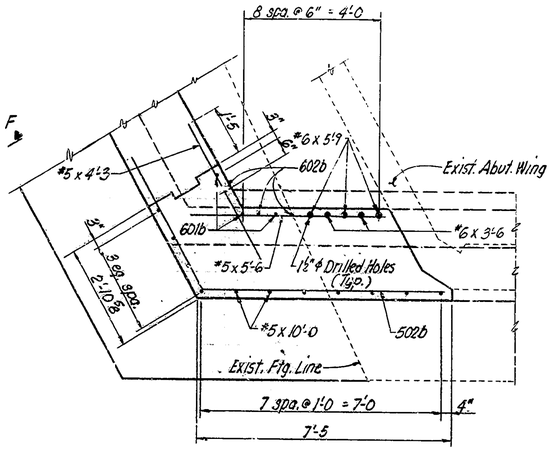
PLAN

Note:
Dimensions and reinforcing steel same as Abut. N°1 except as noted. See Dwg. R3.

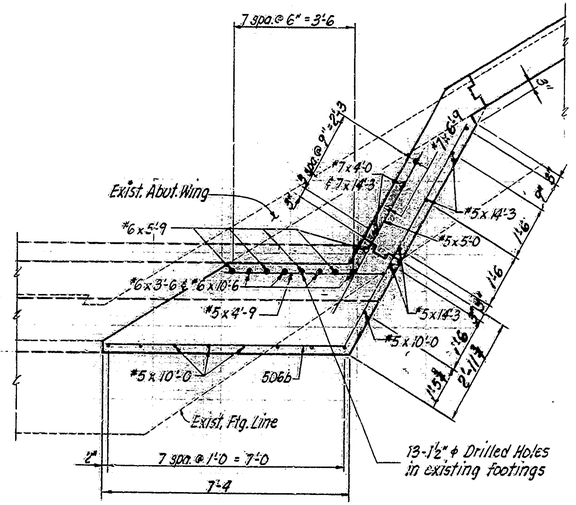


ELEVATION

Note:
See Bridge Plan Sht. for Reinforcing Bar Lists and 4" ℄ Drains.
See Bridge Plan Sht. for Wing ℄ Construction Joints.
See Bridge Plan Sht. for Elevations of Beg. Top Details.
See Dwgs. R3 and R5 for Bill of Materials and additional details.



SECTION K-K
Scale: ½" = 1'-0



SECTION L-L
Scale: ½" = 1'-0

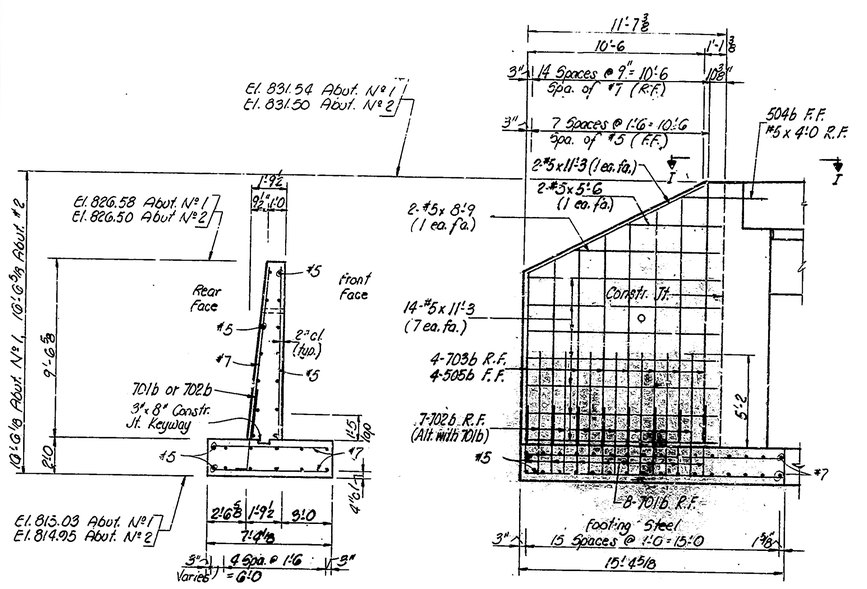
ABUTMENT N° 2 DETAILS

INDIANA DEPARTMENT OF HIGHWAYS

SCALE:- ½" = 1'-0 Unless Noted DATE:- JANUARY 25, 1985
SUBMITTED FOR APPROVAL John S. Spencer
DRAWING:- R4 OF R9 , SHEET:- 6 OF 43
PROJECT:- FR-158-2(5)
CONTRACT NO. B-16317
BRIDGE FILE:- 224-90-15454

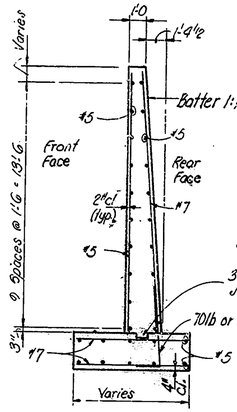
JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

DESIGNED _____ C'K'D _____
DRAWN Mch 8-13-84 C'K'D J.K. 8-23-84
TRACED _____ C'K'D _____

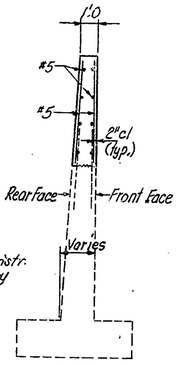


END ELEVATION B-B
Scale: 1/4" = 1'-0

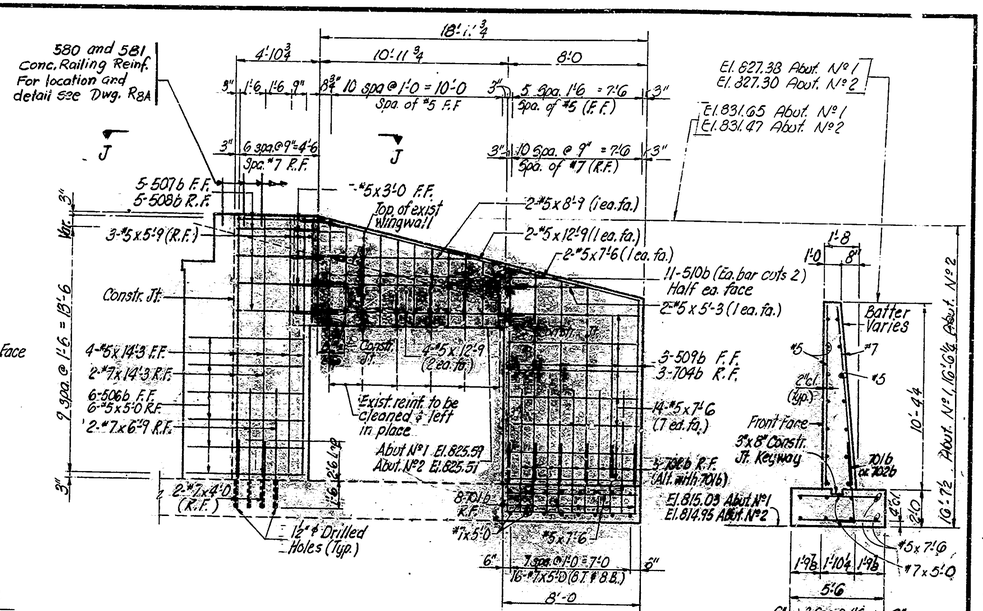
ELEVATION C-C
Scale: 1/4" = 1'-0



SECTION D-D
Scale: 1/4" = 1'-0

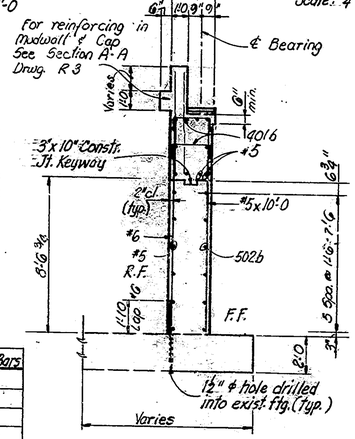


SECTION E-E
Scale: 1/4" = 1'-0

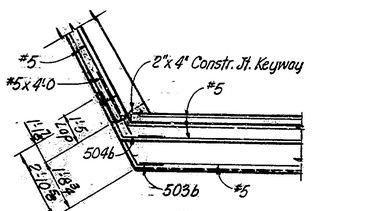


ELEVATION F-F
Scale: 1/4" = 1'-0

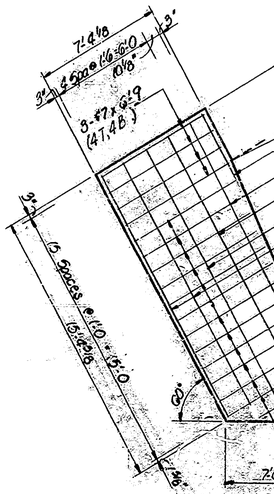
END ELEVATION G-G
Scale: 1/4" = 1'-0



SECTION H-H
Scale: 1/4" = 1'-0

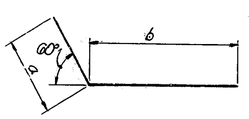


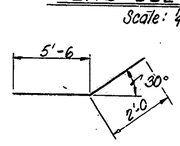
SECTION I-I
Scale: 1/4" = 1'-0



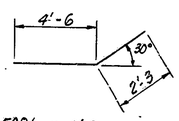
FOOTING REINFORCING WING "B" & "C"
Scale: 1/4" = 1'-0

Mark	a	b	Length
502b	4'-0	7'-0	11'-0
503b	4'-3	1'-6	5'-9
504b	3'-0	3'-0	6'-0
506b	7'-0	7'-0	14'-0



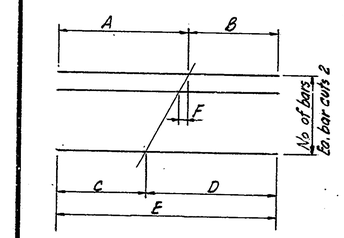


507b x 7'-6



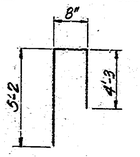
508b x 6'-9

NOTES:
See Bridge Standard C1 for Reinforcing Bar Notes.
See Bridge Standard C3 for Type "A" Constr. Jt.
See Dwg R3 for additional details.

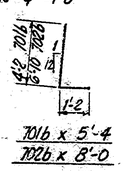


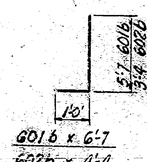
CUTTING DIAGRAM

Mark	A	B	C	D	E	F	No. Bars
703b	9'-4	14'-4	11'-5 3/4	12'-2 1/4	23'-8	8 5/8"	4
704b	10'-2	11'-10	10'-10	11'-2	22'-0	3 3/8"	3
705b	6'-9	5'-10	5'-10	6'-9	12'-7	1"	12
505b	9'-4	14'-4	11'-5 3/4	12'-2 1/4	23'-8	8 5/8"	4
509b	10'-2	11'-10	10'-10	11'-2	22'-0	3 3/8"	3
510b	5'-7	3'-4	3'-4	5'-7	8'-11	2 5/8"	11

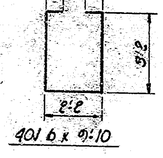


501b x 10'-1

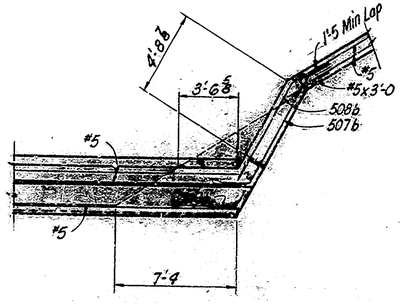




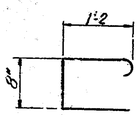
601b x 6'-7
602b x 4'-4



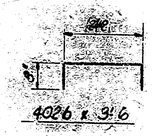
401b x 9'-10



SECTION J-J
Scale: 1/4" = 1'-0



403b x 3'-6



402b x 3'-6

DESIGNED ____ C'K'D ____
DRAWN M.E.B. 8-15-84 C'K'D J.K. 8-23-84
TRACED ____ C'K'D ____

Rev: 2-26-86, Conc. Railing bar shown (See Section FF)

ABUTMENT N° 1 & N° 2 DETAILS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL John S. Spencer

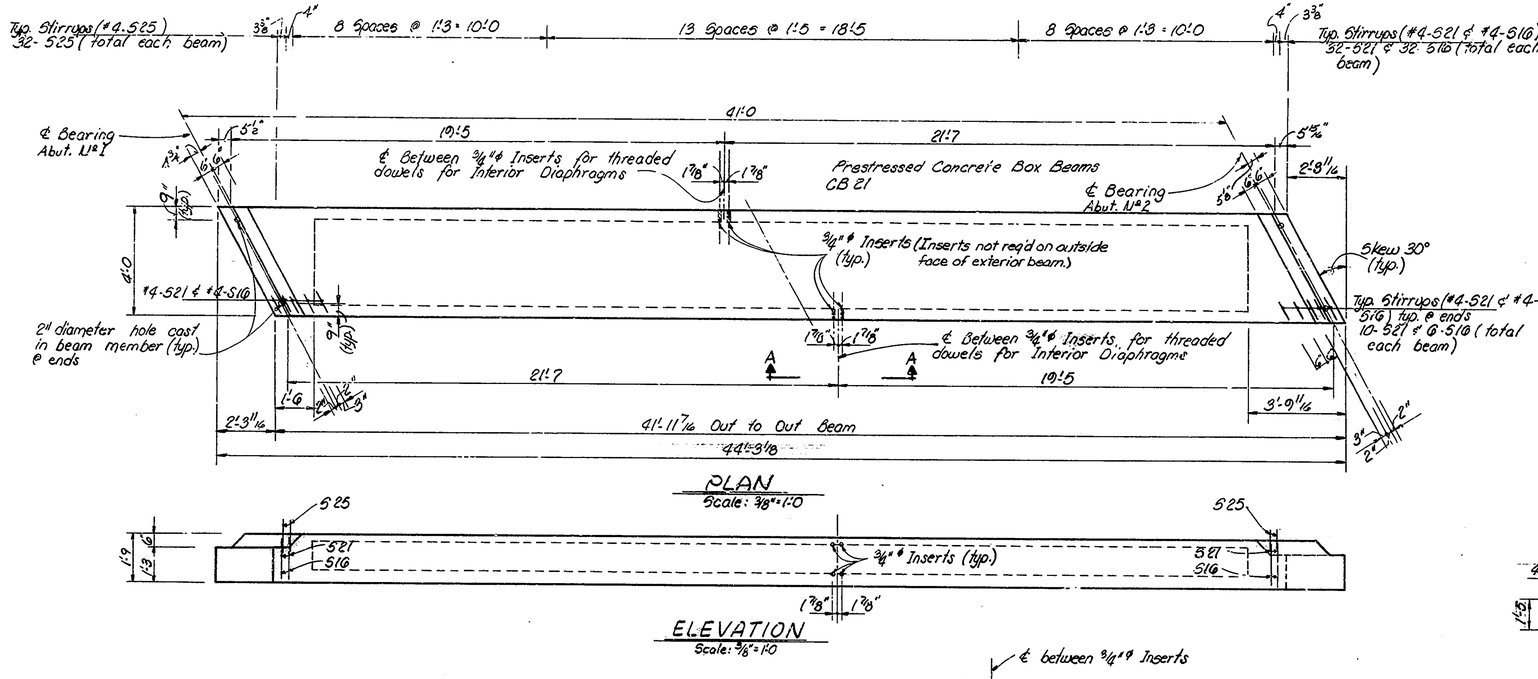
DRAWING:- R5 OF R9 SHEET:- 7 OF 43

PROJECT:- FR-158-2(5)

CONTRACT NO. B-16317

BRIDGE FILE:- 224-90-15454

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER



PRESTRESSING NOTES

Prestressed concrete bridges shall be designed in accordance with 1977 AASHTO Specifications and Interim Specifications.

Minimum 28 day compressive strength of concrete to be 5,000 psi.

Minimum compressive strength of concrete at time of prestressing to be 4,000 psi.

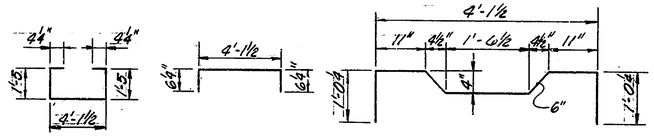
Prestressing steel to be 1/2" diameter 7 wire stress relieved strands with a minimum tensile strength of 270,000 psi.

Beam manufacturer shall equip beams with satisfactory lifting devices and the contractor shall follow the manufacturer's recommendations for handling and erecting.

The material and labor for the manufacture, transportation and erection of prestressed concrete box beams, including elastomeric bearing pads, dowels and grout, inserts and 3/4"ø x 8" hex bolts, lifting devices, and sealer on exterior beams, as specified, shall be included in the lump sum item "Concrete Structural Members."

Sealer required on the outside face of the exterior beams, to be done by the beam fabricator in the shop as shown on detail plans. Do not rub.

The beam manufacturer shall furnish the Engineer, thru the contractor, five (5) sets of shop drawings for his approval prior to the casting of the beams.



516 x 7'-8 | 321 x 5'-2 | 325 x 6'-5

Bridge seat elevations were calculated using design camber and dead load deflection of slab (residual beam camber) with the top of beam at bottom of slab elevation at centerline. Actual cambers which are greater than design cambers will be taken care of by permitting the top of beam to extend into the slab (limit of 1 inch)

Initial Camber =	5/16"
Conc. D.L. Deflection =	-5/16
Net Camber =	0"

NOTE:
For Reinforcing Bar Notes see Br. Std C1
For General Notes see Dwg R2
For Prestressed Box Beam details see Br. Std. PBG, PB9B & PB10
See Dwg R7 for Typical Section thru Interior Diaphragm.

BEAM DETAILS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL John S. Spencer

DRAWING:- R6 OF R9 SHEET:- 8 OF 43

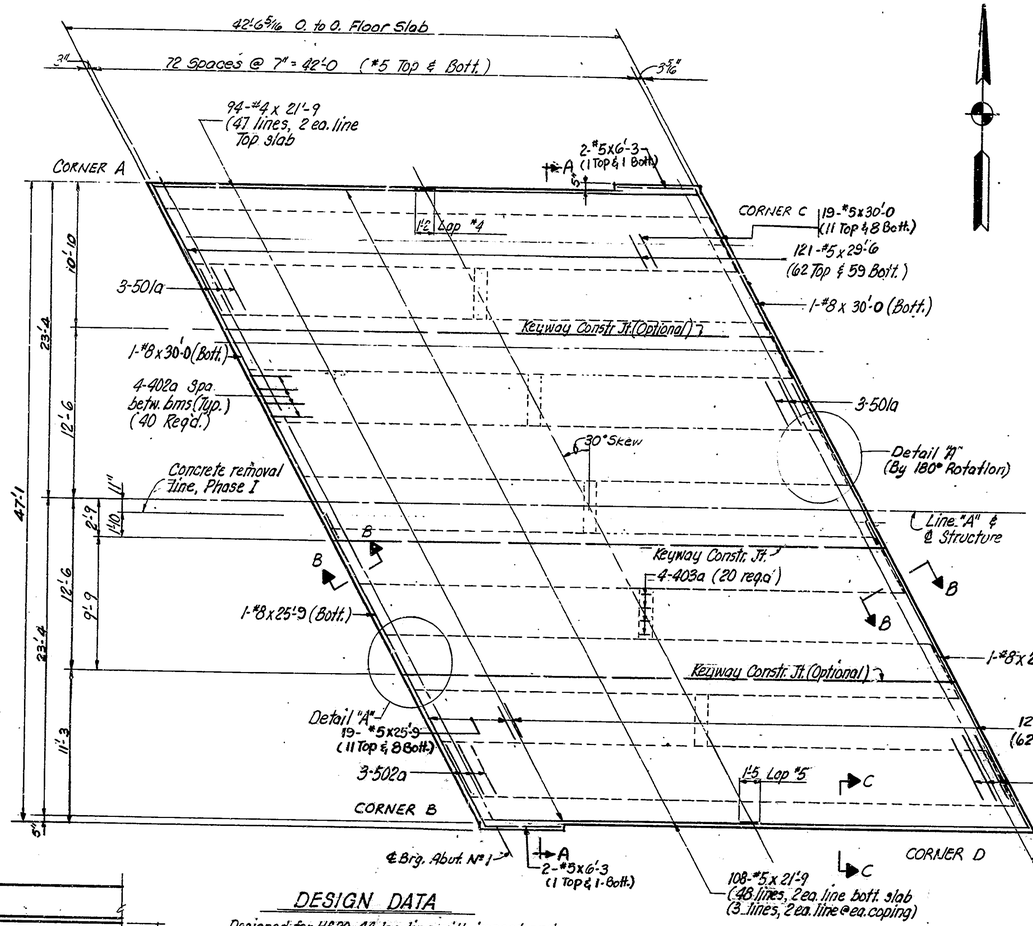
PROJECT:- FR-158-2(5)

CONTRACT NO. B-16317

BRIDGE FILE:- 224-90-15454

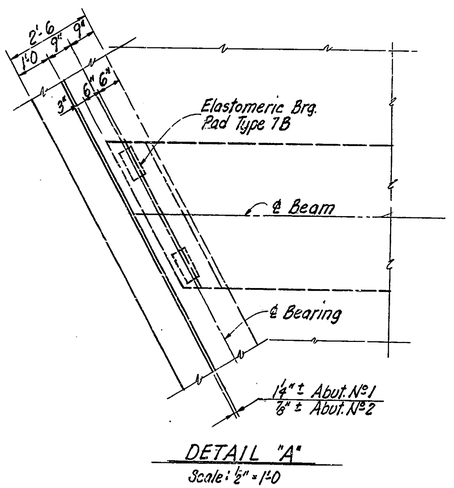
JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

DESIGNED	MH 7-20-84	C'K'D	
DRAWN	MEB 8-15-84	C'K'D	JK 8-23-84
TRACED		C'K'D	



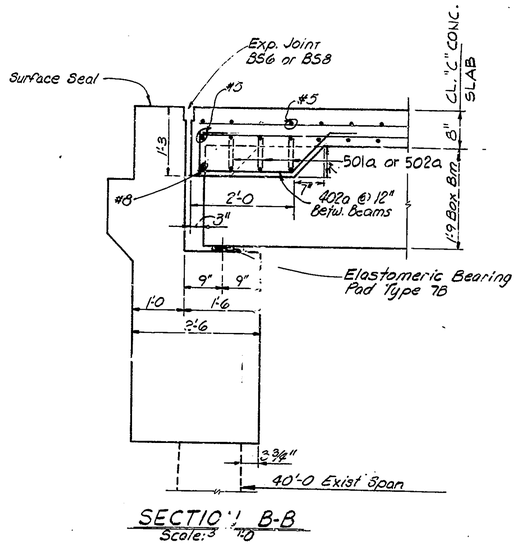
PLAN

Scale: 3/16" = 1'-0



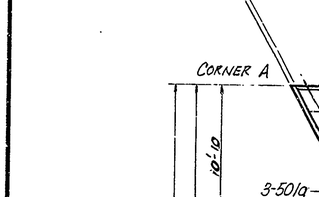
DETAIL "A"

Scale: 1/2" = 1'-0



SECTION B-B

Scale: 3/8" = 1'-0



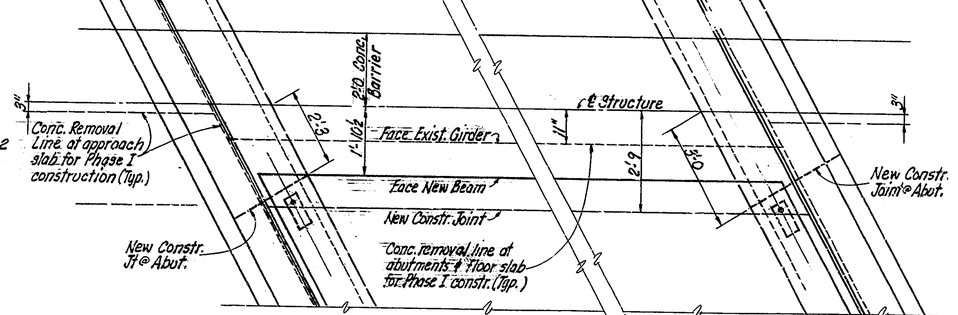
DESIGN DATA

Designed for H520-44 loading with impact and distribution of loads in accordance with 1977 A.A.S.H.T.O. Specifications and Interim Specifications.

Dead Load includes 35 p.s.f. for weight of future wearing surface.

Slab designed with structural depth of 6 1/2".

Poured in place concrete: f_s = 20,000 p.s.i.
f_c = 1,200 p.s.i.

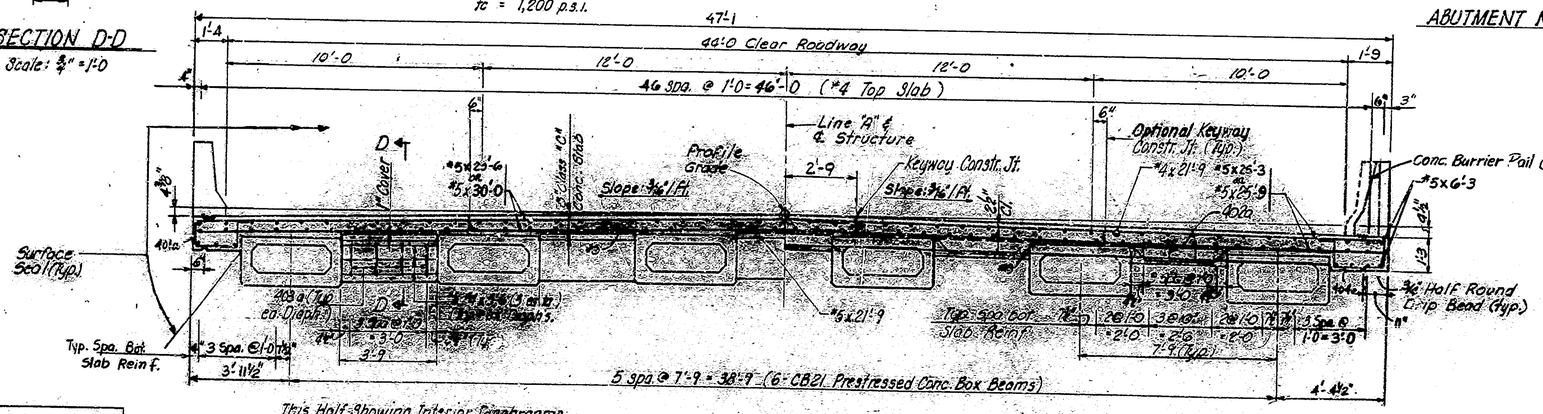


ABUTMENT N° 1

ABUTMENT N° 2

PLAN SHOWING CONCRETE REMOVAL LINES - PHASE I CONSTRUCTION

Scale: 1/2" = 1'-0



This Half Showing Interior Diaphragms

This Half Showing End Diaphragms

SECTION A-A

Scale: 3/8" = 1'-0

NOTE: See Bridge Std. C1 for Reinforcing Bar Notes, for Additional Details See Dwg. R6.

SUPERSTRUCTURE DETAILS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL John L. Spencer

DRAWING:- R7 OF R9 SHEET:- 9 OF 43

PROJECT:- FR-158-2(5)

CONTRACT NO. B-16317

BRIDGE FILE:- 224-90-1545A

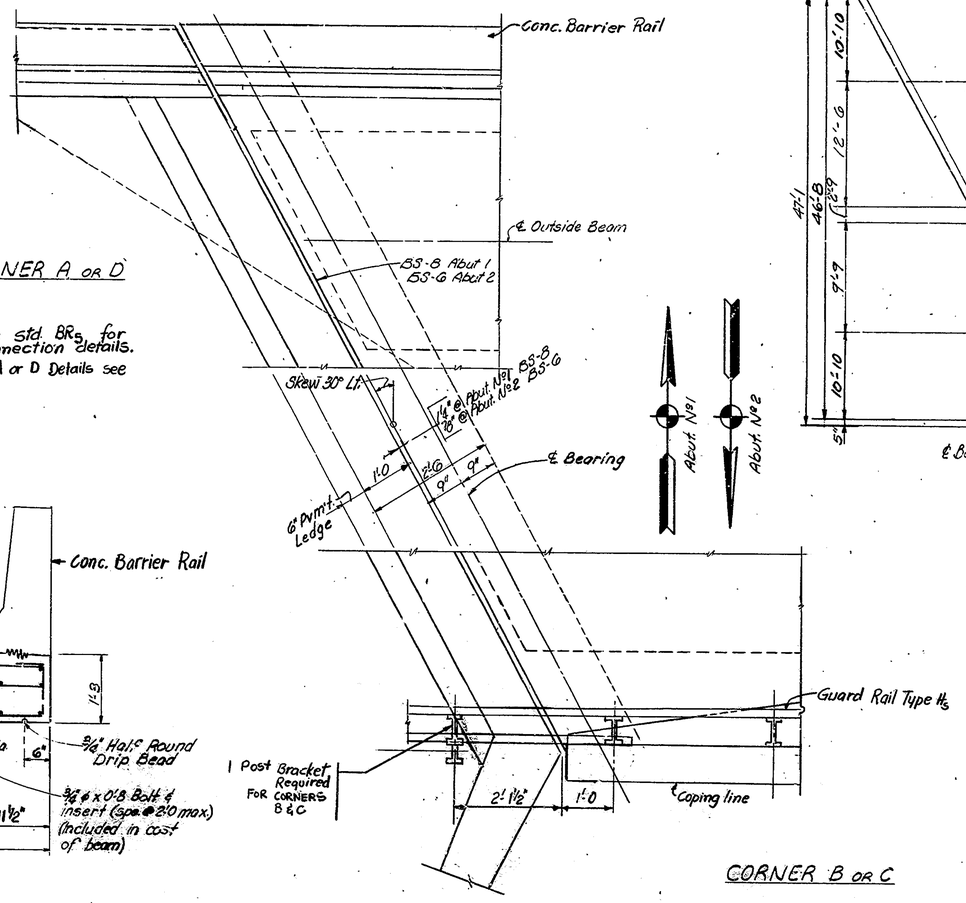
JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

DESIGNED C'K'D

DRAWN MEB 8-20-86 C'K'D JK 8-25-86

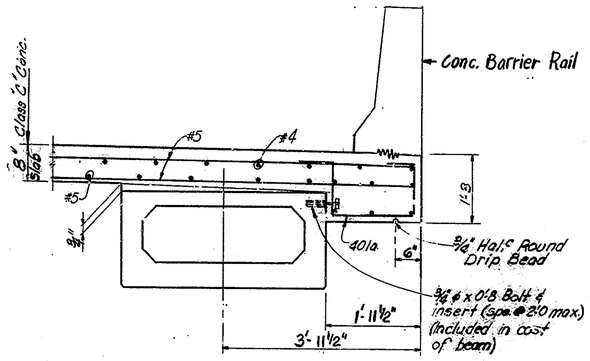
TRACED C'K'D

Rev. 2-26-86 Cut to Cut Coping distance Reinforcing bar length and Spa., Conc. Barrier Rail

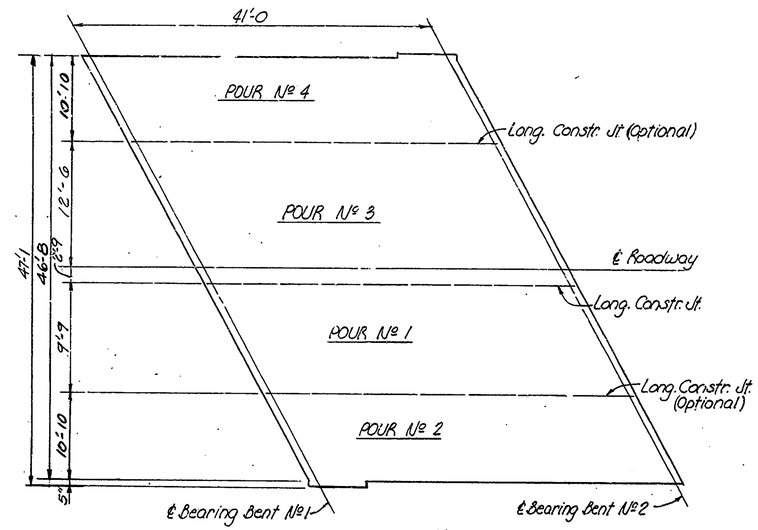


CORNER A OR D

Note: See Bridge Std. BR5 for railing connection details. For Corner A or D Details see Sheet No. 88.



SECTION C-C
Scale: ¾"=1'-0



POUR DIAGRAM
Scale: ⅛"=1'-0

NOTE:
Sequence of pours to be made in order of pour numbers. All superstructure construction joints are optional except as noted, and pours may be made continuous provided the pour terminates at a construction joint indicated on the plans. The contractor may change the width of pours, sequence of pours, or location of construction joints subject to the approval of the Engineer.

The top reinforcing in the deck shall be securely tied down to the deck forms and/or the beams to prevent lifting during concrete placement.

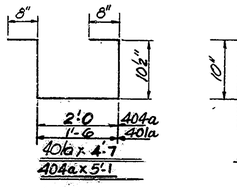
BILL OF MATERIALS SUPERSTRUCTURE

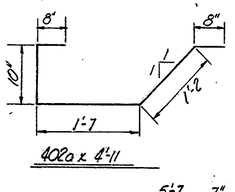
EPOXY COATED REINF. STEEL

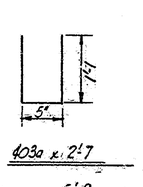
Size & Mark	No. of Bars	Length	Weight (lbs.)
#8	2	30'-0	
#8	2	25'-9	
	Total #8		298
501a	6	31'-2	
502a	6	26'-3	
#5	19	30'-0	
#5	121	29'-6	
#5	19	25'-9	
#5	121	25'-3	
#5	108	21'-9	
#5	4	6'-3	
	Total #5		10250
401a	72	4'-7	
402a	40	4'-11	
403a	20	2'-7	
404a	14	5'-1	
#4	94	21'-9	
#4	30	3'-6	
	Total #4		1870
Total Epoxy Coated Reinf. Steel			13018

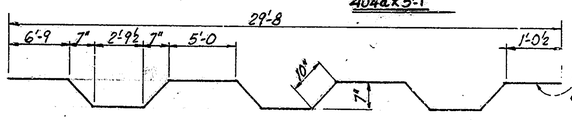
CONCRETE	
Class "C" in Superstructure	
Pour No. 1	11.1 cys.
Pour No. 2	14.1 cys.
Pour No. 3	17.4 cys.
Pour No. 4	13.1 cys.
Interior Diaphragms	1.7 cys.
Total Class "C"	57.4 cys.
MISCELLANEOUS	
Surface Seal	2143 Sft.
Exp. Jt. Type B56	54 Lft.
Exp. Jt. Type B58	54 Lft.

CORNER B OR C

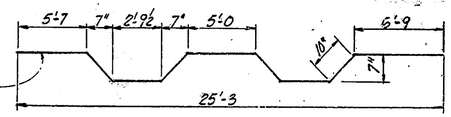






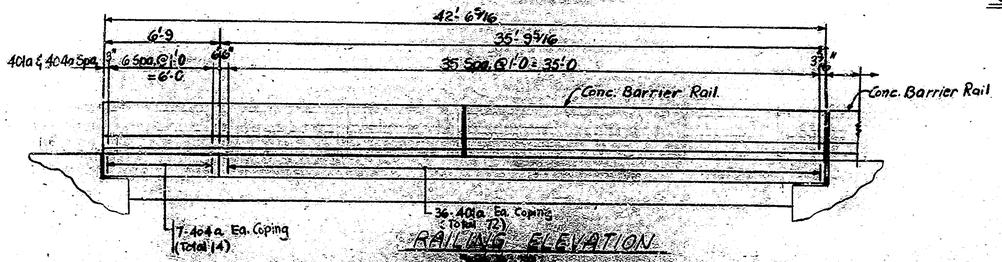


501a x 31'-2



502a x 26'-3

CORNER DETAILS
Scale: ¼"=1'-0



RAILING ELEVATION

NOTE:
See Bridge Std. C1 for Reinforcing Bar Notes. For Additional Details See Dwg. R7.

Rev. 2-27-86, corner Detail, Railing Detail, Bill of Materials, Billing Diagram, Sect. C-C, Pour Diagram.

SUPERSTRUCTURE DETAILS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL John S. Spencer

DRAWING:- R8 OF A9 SHEET:- 10 OF 45

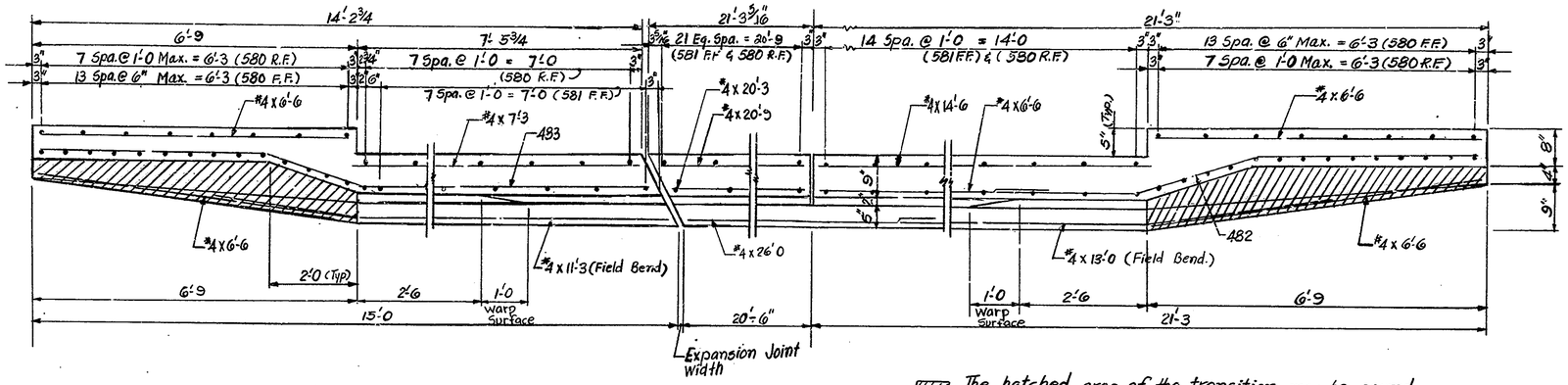
PROJECT:- FR-158-2(5)

CONTRACT NO. B-16317

BRIDGE FILE:- 284-90-1545A

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

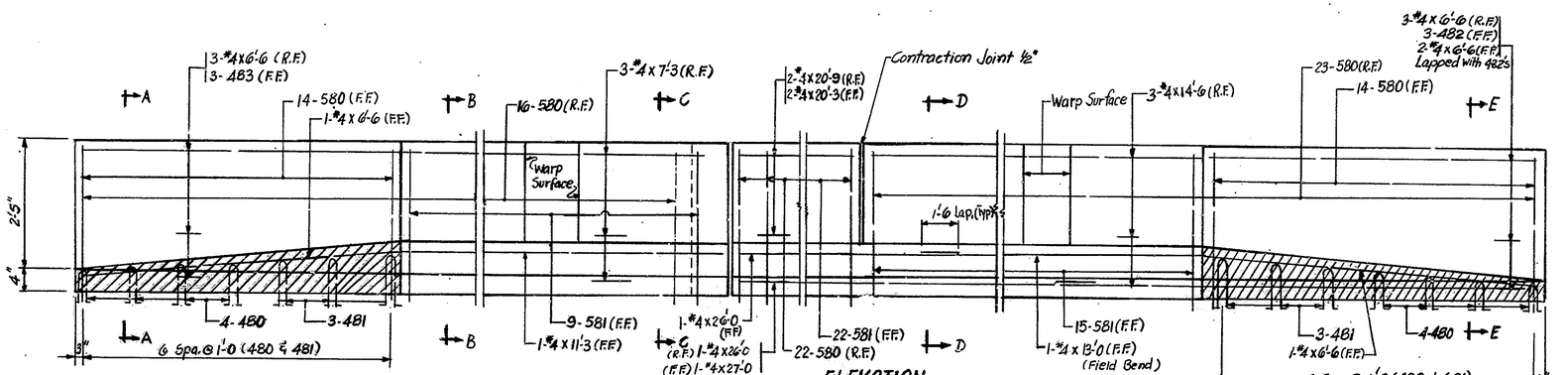
DESIGNED C'K'D
DRAWN MEB 8-21-84 C'K'D JK 8-23-84
TRACED C'K'D



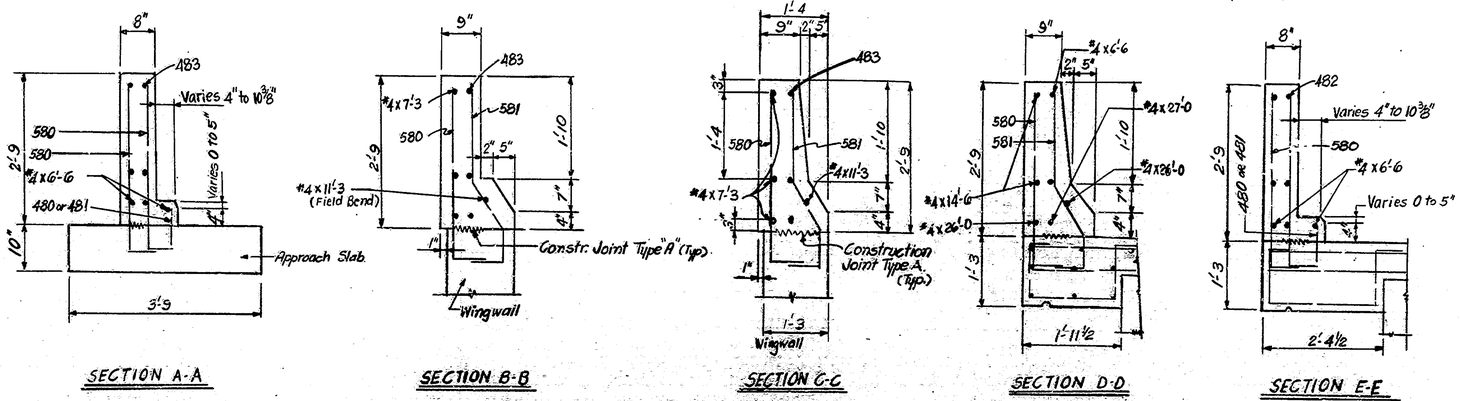
The hatched area of the transition may be poured either as a curb or monolithic with the railing connection.

PLAN

NORTH SIDE SHOWN
SOUTH SIDE SAME BY 180° ROTATION



ELEVATION



SECTION A-A

SECTION B-B

SECTION C-C

SECTION D-D

SECTION E-E

Notes:

For Reinforcing Bar Notes and bar bending details, see Bridge Std. C.
For Guard Rail connection details, see Bridge Std. BR.
For Bill of Materials and other details, see Sheet No. 10B
For Constr. Joint Type "A" details see Br. Std. C3.

CONCRETE BARRIER RAIL DETAILS

INDIANA DEPARTMENT OF HIGHWAYS

SCALE:- NO SCALE DATE:- 19

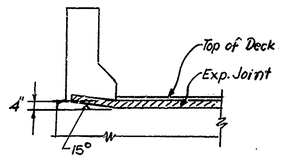
SENIOR DESIGNER

DRAWING:- R6A OF R9 SHEET:- 10A OF 43
PROJECT:- FR-158-2(5) STATION:-
BRIDGE CONTRACT NO. B-16317
BRIDGE FILE:- 224-90-1545A

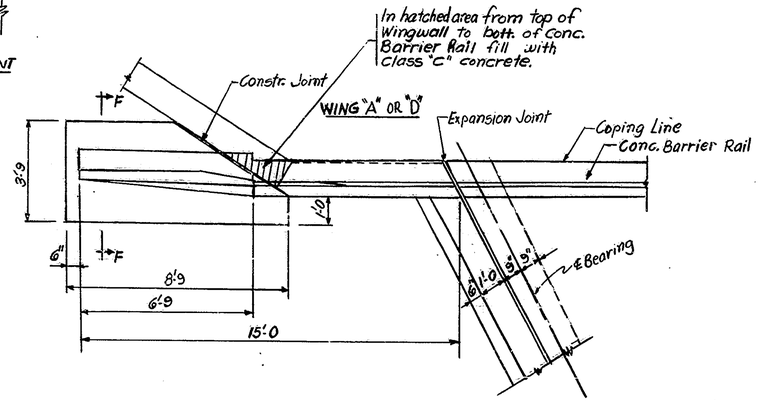
DESIGNED C'K'D
DRAWN S.J.P. 2/21/86 C'K'D F.E.S. 4/21/86
TRACED C'K'D

SF-22317

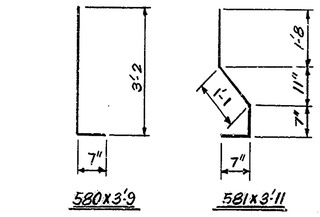
TP118

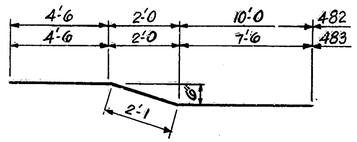


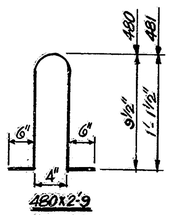
SECTION SHOWING EXP. JOINT IN BARRIER RAIL
No Scale



DETAIL @ ABUTMENT #1
(Same By 180° Rotation @ Abut. #2)
Scale: 3/8"=1'-0







BILL OF MATERIALS

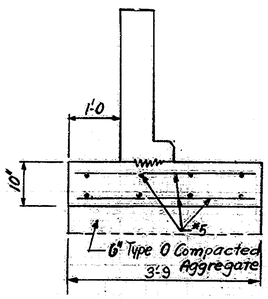
PAD @ ABUTMENT #1
SAME FOR ABUTMENT #2

REINFORCING STEEL			
SIZE & MARK	NO. OF BARS	LENGTH	WEIGHT (LBS.)
#5	3	11'-6	
#5	2	8'-5	
#5	5	4'-0	
#5	8	3'-5	
	Total #5	Steel =	103
	CONCRETE		
10" Reinf. Conc. Pavmt.			2.9 Sys

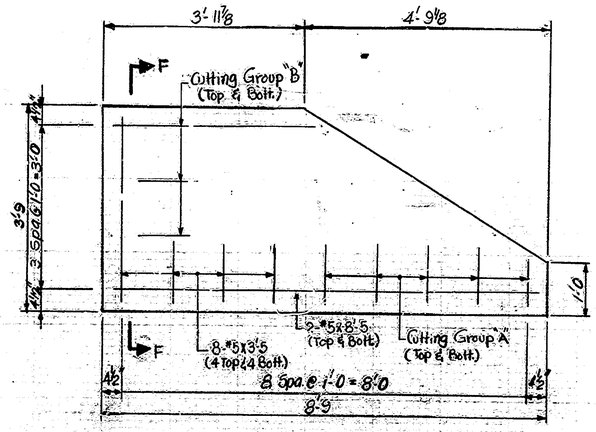
BILL OF MATERIALS

CONC. BARRIER RAILING
(FOR ONE SIDE - 10 REQUIRED)

EPOXY COATED REINFORCING STEEL			
SIZE & MARK	NO. OF BARS	LENGTH	WEIGHT (LBS.)
580	89	3'-9	
581	46	3'-11	
	Total	#5	536
480	8	2'-9	
481	6	3'-5	
482	3	16'-7	
483	3	14'-1	
#4	1	27'-0	
#4	2	26'-0	
#4	2	20'-9	
#4	2	20'-3	
#4	3	14'-6	
#4	1	13'-0	
#4	1	11'-3	
#4	3	7'-3	
#4	10	6'-6	
	Total	#4	301
Total Epoxy Coated Reinf.			837
MISCELLANEOUS			
Surface Seal			3625Ft.
CONCRETE			
CONCRETE CL. "C" IN RAILING			5.5 Cys

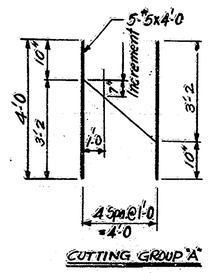


SECTION FF

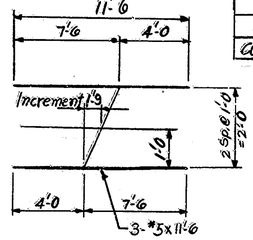


PAD DETAIL
Scale: 3/4"=1'-0

PLAN



CUTTING GROUP "A"



NOTES:
See Bridge Std. C, for Reinforcing Bar Notes.
For additional Details see sheet No. R8A.

CONCRETE BARRIER RAIL PAD DETAILS AND BILL OF MATERIALS

INDIANA DEPARTMENT OF HIGHWAYS

SCALE:- AS NOTED DATE:- 19

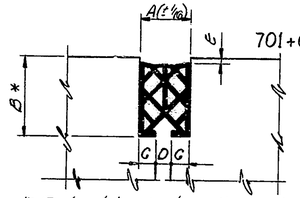
SENIOR DESIGNER

DRAWING:- R8B OF R9 SHEET:- 10B OF 43
PROJECT:- FR-158-2(5) STATION:-
BRIDGE CONTRACT NO. B-16317
BRIDGE FILE:- 224-90-1545A

DESIGNED ____ CK'D ____
DRAWN SWP 2/2/86 CK'D FES 4/2/86
TRACED ____ CK'D ____

SF-22317

TP118



* To be determined in the field See the Special Provisions.

Br. Seal	A	B	C	D	E
BS G	1 5/8	*	3/8"	7/8"±	1/2"
BS B	2"	*	3/8"	1 1/8"±	1/2"

DETAIL OF EXP. JOINT TYPE BS

701+00 702+00 703+00 704+00

SEC. 32, T28N, R13E
JEFFERSON TWP.
WELLS CO.

SEC. 5, T27N, R13E
LANCASTER TWP.
WELLS CO.

(Lawn) (Soybeans) (Corn)

BEGIN PROJECT Nº FR-158-2( )
Sta. 701+70.0 LINE "A"

END PROJECT Nº FR-158-2( )
Sta. 705+85.00

Remove 41 Lft. Exist. Guard Rail & Install 100 Lft. Guard Rail Class H$_2$ With GRET Type I

Remove and Replace existing R.C. Bridge Appr. (Typ.)

Remove 41 Lft. Exist. Guard Rail & Install 100 Lft. Guard Rail Class H$_2$ with GRET Type I

NOTE: GRET = Guard Rail End Treatment

App. ℄

Field Ent. (Dirt)

4' Paved Side Ditch

DITCH

EIGHT MILE

Bitum. Shoulder (Typ.)
Bitum. Widening (Typ.)
Bituminous Wedge and Level
90'-0"
30'-0"
55°45'
Bitum. Widening (Typ.)
Bitum. Shoulder (Typ.)
24'-0
11'-0
11'-0
8"
8"

U.S. #224

699+95.0
704+21.13
702+12.69
702+05.70
705+04.30
705+97.32
707+45.0
705+85.0

Bitum. Widening
Line "A"
+35
+38.87

6" Gas Line

50'-0

G.R.E.T. Type I

Str. Nº 10

Str. Nº 11

Taper to existing shoulder line in 50' (typ.)

Taper to existing shoulder line in 50' (typ.)

Install 38 Lft. Guard Rail Class H$_2$ with GRET Type I Includes 25 Lft. of stop curved guard rail.

Remove 15 Lft. Exist. Guard Rail & Install 28 Lft. Guard Rail Class H$_2$ Includes 18 Lft. of stop curved guard rail

W-Beam Guard Rail Terminal Section (2 Req'd)

Special Class II Drive
Sta. 703+13 Rt.
W= 12'-03

Ditch (Grass)

CONSTRUCTION PROCEDURE

1. Install traffic control devices to divert traffic for Phase I Construction as shown on Sheet Nº 2.
2. Remove one half superstructure, R.C. Girder, portions of substructure and bridge approach slab at each end.
3. Extend abutments, place new beams and rebuild mudwalls as required.
4. Pour concrete superstructure, bridge approach slab, and construct concrete pads and Barrier Railing.
5. Reshape shoulders, place bituminous wedging material, and all other work shown on the plans, including the installation of guard rail.
6. Reset traffic control devices to divert traffic to completed lane and repeat steps 2 thru 5.

The numbers do not necessarily indicate the sequence of operations.

"All removal equipment used for partial concrete removals of Bridge Floors shall be Handheld. Pneumatic hammers, 30 lbs. maximum weight, shall be used for all Areas of Bridge Deck Patching and all areas within 24 inches of full depth removal lines. Pneumatic hammers, up to 90 lbs. maximum weight, may be used for all other removals outside these limits. Deck areas to be removed full depth shall be completely separated (Concrete and Steel) from adjacent concrete before hammers heavier than 30 lbs. may be used.

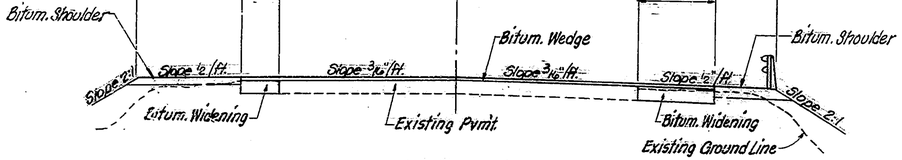
BILL OF MATERIALS APPROACHES

REINFORCING STEEL

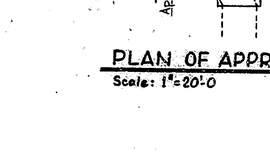
Size or Mark	Nº of Bars	Length	Weight (Lbs.)
551	96	20'-6	
#5	22	16'-9	
#5	22	11'-10	
Total Reinf. Steel			2708

MISCELLANEOUS

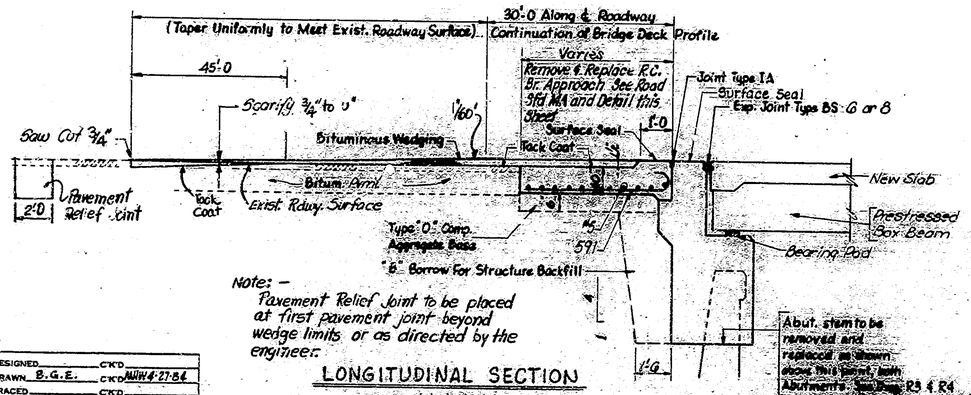
R.C. Pavement (10 inch)	146 Sys.
Type "O" Comp. Aggregate	49 Tons
"B" Borrow for Structure Backfill	109 Cu.
Surface Seal	55 Sq.



SECTION A-A
Not to Scale

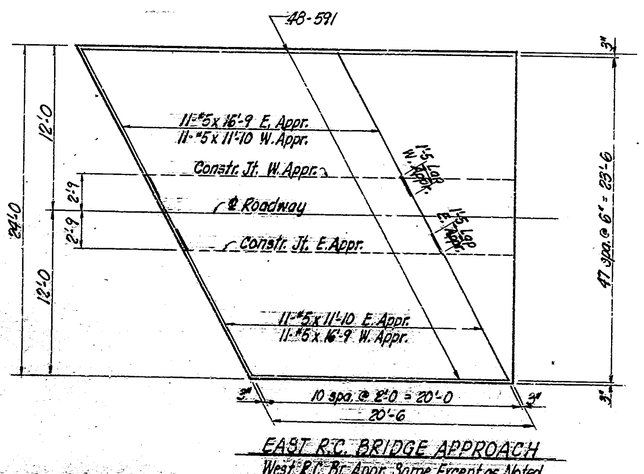


PLAN OF APPROACHES
Scale: 1"=20'-0



Note: — Pavement Relief Joint to be placed at first pavement joint beyond wedge limits or as directed by the engineer.

LONGITUDINAL SECTION
Not to Scale



EAST R.C. BRIDGE APPROACH
West R.C. Br. Appr. Same Except as Noted
Scale: 3/16" = 1'-0

APPROACH DETAILS

INDIANA STATE HIGHWAY COMMISSION

SCALE:- As Noted DATE:- JANUARY 25, 1985

SUBMITTED FOR APPROVAL

DRAWING:- A9 OF A9 SHEET:- 11 OF 43

PROJECT:- FR-158-2(5)

CONTRACT NO. 16317

BRIDGE FILE:- 224-90-1545A

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

DESIGNED ____ C'K'D ____
DRAWN B.G.E. C'K'D MJH 4-27-84
TRACED ____ C'K'D ____

Field Notes: See Book BRMA pp. 1-19.

Rev: 3-14-86, Bituminous Wedge and Level Detail. Delete Guard Rail Type G. Add Guard Rail Type H

Rev: 1-14-87, Relocate Drive @ Sta. 703+13 & Change G.R. Quantities.

6-17-85 - Drive @ Sta. 703+26 Rt: Change Class II to Special Class II Drive, Change Width

7-11-85 - Delete Class II Drives @ Sta. 705+30 Lt & 705+55 Rt

12-16-85 - Delete Special Class II Drive @ Sta. 703+26 Rt., Added Special Class II Drive @ Sta. 703+10 Rt.

BRIDGE FILE 224-90-1545A

STRUCTURE QUANTITIES

ITEM	CONCRETE CLASS C SUPERSTR CU. YDS.	CONCRETE CLASS A SUBSTR CU. YDS	CONCRETE CLASS B ABOVE FTG. CU. YDS.	CONCRETE CLASS B IN FTG. CU. YDS.	CONCRETE RAILING CLASS C CU. YDS.	CONCRETE RAILING CLASS C LIN. FT.	REINF. STEEL TOTAL LBS.	STRUCT. STEEL *** LBS.	EPOXY COATED REINF. STEEL LBS.	ANCHOR PLATES M-AP EACH	PILES UNTREATED TIMBER NO.	PILES UNTREATED TIMBER LIN. FT.	PILES TREATED TIMBER NO.	PILES TREATED TIMBER LIN. FT.	PILES STEEL ENCASED CONC. NO.	PILES STEEL ENCASED CONC. LIN. FT.	PILES STEEL H BEARING NO.	PILES STEEL H BEARING LIN. FT.	CAST IRON DRAIN PIPE LBS.	CAST IRON, GRATES, BASINS, & FITTINGS LBS.	RAILING TYPE 5A OR LIN. FT.	EXP. JOINT TYPE B56 LIN.FT.	EXP. JOINT CLASS B58 LIN.FT.	CONC. STR. MEMBERS BOX BEAMS TYPE CB21 SQ. FT.	CONC. STR. MEMBERS I BEAM TYPE LIN.FT.	BRIDGE DECK MEMBRANE SQ.FT.	BITUM MIXTURE FOR APPROACHES TONS	BRIDGE DECK SURFACE SQ. YDS	DECK DRAIN EACH	SURFACE SEAL SQ.FT.	FINISHING AND CURING SQ. YD.	BLASTING AND CLEANING SQ. YD.
ABUTMENT Nº 1		55.2		12.1			4520		668																					109		
ABUTMENT Nº 2		55.2		12.1			4520		668																					109		
SUPERSTRUCTURE	57.4								13,018													54	54	1012						2143		
CONC. BARRIER RAIL					11.0				1674																					724		
Reinf. Steel for Pad							206																									
Reinf. Steel for Approach Structures																																
Reinf. Steel for R.C. Bridge Approaches							2708																									
Reinf. Steel for Lip Gutter, Pvmt., Tapers, etc.																														55		
TOTALS	57.4	110.4		24.2	11.0		11954		16028													54	54	1012						3140*		

* Estimated quantity. See the Special Provisions

APPROACH STRUCTURES

STRUCT. NO.	LOCATION	DESCRIPTION SIZE	DESCRIPTION KIND	DESCRIPTION LENGTH LIN. FT.	CONCR. CL. A IN STRS. CU. YDS.	REINF. STEEL LBS.	PIPE END SEC. EACH		REMARKS
	03			~~38~~					
10	STA. 703+13 36' RT.	18"	GROUP "D" (C.S.=.079")	~~38~~			2		REMOVE EXISTING PIPE
11	STA. 705+55 42' RT.	18"	GROUP "D" (C.S.=.064")	185			2		REMOVE EXISTING PIPE
			TOTALS				4		Total of Reinforcing Steel Carried to "Structure Quantities"

PAVED SIDE DITCH & SODDING SUMMARY

LT. OR RT.	STATION TO STATION	PAVED SIDE DITCH (LIN.FT.) TYPE	PAY LENGTH	NO. OF LUGS	PAY LENGTH	CUT OFF WALLS	PAY LENGTH	TOTAL PAY LENGTH	FOR PSD	SODDING (SQ.YD.) FOR DITCHES	SHOULDERS	OTHER	TOTAL SOD
LT.	702+00 TO 703+20	"B"	120	—	—	2	16	136	36				36
RT.	702+00 TO 703+20									73			73
	@ 4 CORNERS OF BR.											82	82
												TOTAL =	191

APPROACH TABLE

LOCATION LT/RT	LOCATION STATION	DESCRIPTION	WIDTH FT.	RADII FT.	GRADE %	LENGTH FT.	DIST. RETURNED M/S FT.	EXCAVATION(CYS) CUT	EXCAVATION(CYS) FILL	BITUM. SURFACE #/SQ.YD.	BITUM. SURFACE TONS	BITUM. BINDER #/SQ.YD.	BITUM. BINDER TONS	BITUM. #/SQ.YD	BASE TONS	COMPR AGG. BASE Depth (in)	COMPR AGG. BASE TONS
RT.	703+13	Special Class ~~X~~ Drive	12	9'-20'	~~10~~	40			18	110	~~2~~			220	~~4~~	~~5~~	~~7~~
	03			15'-25'	4.17	120					8				16		27

NOVEMBER 1978

SUMMARIZED MB 9-11-84 C'K'D MHW 9-12-84

TRACED _____ C'K'D _____

NOTES:
Weight of Spirals includes weight of 1½ extra turns top and bottom.
Spacers and 1½ turns at tops included in cost of Spiral.
*** The weight of structural steel is approximate only, and it shall be the Contractor's responsibility to determine the weight on which he bases his bid.
For Test Bor Samples See Bridge Standard C1.

REVISIONS

DATE	ITEM
6-17-85	Structure No. 10, Special Class II Drive
7-1-85	Delete Class II Drives @ Sta 705+20 Lt & Sta 706+55 Rt
12-16-85	Delete Spl. Cl. II Drive @ Sta. 703+26 R Added Spl Cl. II Drive @ Sta 703+13 Rt.
3-13-86	Class C is Superstr. Epoxy Coated Reinf. Steel in Superstr. Surface Seal, Add Quant. for Epoxy Coated Reinf. Steel. Surface Seal for Conc. Barrier Rail, Add Reinf. Steel for Pad., Delete Railing Type 5A.
1-14-87	Special Class II Drive Lengthened, Quantities Adj.

BRIDGE SUMMARY
INDIANA STATE HIGHWAY COMMISSION

DATE JANUARY 25, 1985

SUBMITTED FOR APPROVAL: John S. Spencer

SHEET 12 OF 43

PROJECT: FR-158-2(5)
CONTRACT NO: B-16317
BRIDGE FILE: 224-90-1545A

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER

TP103

ESTIMATE OF QUANTITIES

STRUCTURE PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
51002	CONCRETE CLASS C IN SUPERSTRUCTURE	CYS.			57.4
51001	CONCRETE, CLASS A IN SUPERSTRUCTURE	CYS.			
51005	CONCRETE, CLASS A IN SUBSTRUCTURE	CYS.			110.4
51010	CONCRETE, CLASS B ABOVE FOOTINGS	CYS.			
51015	CONCRETE, CLASS B IN FOOTINGS	CYS.			24.2
51875	SPECIAL CLASS A CONCRETE	SFT			
51045	CONCRETE STRUCTURAL MEMBERS	LSUM			1
51030	REINFORCING STEEL	LBS.			11954
51035	STRUCTURAL STEEL	LBS.			
51038	STRUCTURAL STEEL	LSUM			
51032	EPOXY COATED REINF STEEL	LBS.			16028
51090	BRONZE PLATES	LBS.			
51050	ANCHOR RODS (MK-AR 1)	EACH			
51055	ANCHOR RODS (MK-AR 2)	EACH			
51060	ANCHOR RODS (MK-AR 3)	EACH			
51065	ANCHOR RODS (MK-AR 4)	EACH			
51070	ANCHOR PLATES (MK-AP 1)	EACH			
51075	ANCHOR PLATES (MK-AP 2)	EACH			
51080	ANCHOR PLATES (MK-AP 3)	EACH			
51085	ANCHOR PLATES (MK-AP 4)	EACH			
51112	ANCHOR BOLTS	EACH			
51068	TIE DOWN ASSEMBLY MK-UA	EACH			
51095	CAST IRON, DRAIN PIPE, 4 INCH	LBS.			
51100	CAST IRON, DRAIN PIPE, 6 INCH	LBS.			
51105	CAST IRON, DRAIN PIPE, 8 INCH	LBS.			
51110	CAST IRON, GRATES, BASINS AND FITTINGS	LBS.			
51134	REMOVAL OF PRESENT RAILING	LFT.			
51132	RAILING RESET	LFT.			
51115	RAILING (TYPE 5 OR C)	LFT.			
51120	RAILING (TYPE 5A OR C1)	LFT.			
51125	RAILING (TYPE 6 OR D)	LFT.			
51130	RAILING (TYPE 7 OR E)	LFT.			
51020	CLASS C, CONCRETE RAILING	CYS.			11.0
51025	CLASS C, CONCRETE RAILING	LFT.			
51131	BARRIER RAILING TYPE X	LFT.			
51215	CLASS X EXCAVATION	CYS.			
51220	WET EXCAVATION	CYS.			136
51223	WATERWAY EXCAVATION	CYS.			
51224	WATERWAY EXCAVATION	LSUM			
51225	DRY EXCAVATION	CYS.			304
51230	FOUNDATION EXCAVATION (UNCLASSIFIED)	CYS.			
51231	FOUNDATION EXCAVATION (UNCLASSIFIED)	LSUM			
51813	PNEUMATICALLY PLACED MORTAR	SFT.			
51870	REPOINTING MASONRY IN STR'S.	SFT.			50
51814	WELDED STEEL WIRE FABRIC	SFT.			
51859	PAINTING OLD STEEL BRIDGE	LSUM			
51881	EXPANSION JOINT, TYPE BS2	LFT.			
51895	EXPANSION JOINT, TYPE BS6	LFT.			54
51887	EXPANSION JOINT, TYPE BS8	LFT.			54
51888	EXPANSION JOINT, TYPE BS9	LFT.			
51890	EXPANSION JOINT, TYPE BS11	LFT.			
51919	EXPANSION JOINT, CLASS I	LFT.			
51920	EXPANSION JOINT, CLASS II	LFT.			
51921	EXPANSION JOINT, CLASS III	LFT.			
51922	EXPANSION JOINT, CLASS IV	LFT.			

STRUCTURE PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
51135	TIMBER PILES FURNISHED, UNTREATED	LFT.			
51140	TIMBER PILES DRIVEN, UNTREATED	LFT.			
51145	TIMBER PILES FURNISHED, TREATED	LFT.			
51150	TIMBER PILES DRIVEN, TREATED	LFT.			
51155	PILE SHELLS FURNISHED AND DRIVEN (12 INCH)	LFT.			
51160	PILE SHELLS FURNISHED AND DRIVEN (14 INCH)	LFT.			
51185	STEEL H PILES FURNISHED AND DRIVEN (8 BP 36)	LFT.			
51190	STEEL H PILES FURNISHED AND DRIVEN (10 BP 42)	LFT.			
51195	STEEL H PILES FURNISHED AND DRIVEN (12 BP 53)	LFT.			
51210	PILE ENCASEMENT (CONCRETE)	LFT.			
51328	REMOVAL OF PRESENT STRUCTURE (PORTIONS)	LSUM			
51330	REMOVAL OF PRESENT STRUCTURE	LSUM			1
51335	TEMPORARY BRIDGE AND APPROACHES	LSUM			
51366	CONCRETE SLOPEWALL 5 INCH	SYS.			
51365	SLOPEWALL	SYS.			
51370	RIPRAP	SYS.			
51375	REVETMENT RIPRAP	TON			
51371	HANDLAID RIPRAP 12 INCH	SYS.			
51372	DUMPED RIPRAP	TON			
51106	DECK DRAINS	EACH			
51395	STEEL DRAIN PIPE (6 INCH)	LSUM			
51400	STEEL DRAIN PIPE (8 INCH)	LSUM			
51092	STEEL PIPE CONDUIT (2 INCH)	LFT.			
51886	RIVETS REMOVED	EACH			
51864	FIELD DRILLED HOLES	EACH			
51867	STRUCTURAL STEEL CUTTING	SIN			
51869	FIELD DRILLED HOLES IN CONCRETE	EACH			36
51821	SURFACE SEAL	LSUM			1
51830	SHEET APPLIED MEMBRANE	LSUM			
51828	LIQUID APPLIED MEMBRANE	LSUM			
51831	MODIFIED PORTLAND CEMENT CONCRETE OVERLAY	CYS.			
51839	MODIFIED PORTLAND CEMENT CONCRETE (SURFACE)	CYS.			
51833	CONCRETE SCARIFYING	SYS.			240
51834	REMOVAL OF SCARIFYING DUST	LSUM			
51835	HANDCHIPPING AND CLEANING	SYS.			
	BLASTING AND CLEANING	SYS.			
51838	FINISHING AND CURING	SYS.			

APPROACH PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
02020	UNCLASSIFIED EXCAVATION	CYS.			
52240	COMMON EXCAVATION	CYS.			62
52245	BORROW	CYS.			490
52250	B BORROW	CYS.			
52255	B BORROW FOR STRUCTURE BACKFILL	CYS.			
52303	REMOVAL OF PAVEMENT	SYS.			118
02235	BREAKING PAVEMENT	SYS.			124
52490	TERMINAL JOINT	LFT.			
52495	CONTRACTION JOINT, TYPE D-1	LFT.			
52280	CONCRETE PAVEMENT REINFORCED (7 INCH)	SYS.			
52285	CONCRETE PAVEMENT REINFORCED (8 INCH)	SYS.			
52290	CONCRETE PAVEMENT REINFORCED (9 INCH)	SYS.			
52300	CONCRETE PAVEMENT REINFORCED (10 INCH)	SYS.			152
06070	CONCRETE SIDEWALK	SYS.			
52305	TYPE P COMPACTED AGGREGATE FOR BASE (SIZE NO. 53)	TON			
52600	COVER AGGREGATE	TON			
52606	COVER AGGREGATE (SIZE NO. 12)	TON			
52605	AGGREGATE FOR SHOULDER DRAINS	TON			
52610	AGGREGATE FOR UNDER DRAINS	CYS.			
52308	TYPE O COMPACTED AGGREGATE FOR BASE (SIZE NO. 53)	TON			9565
52310	SUBBASE	CYS.			
52315	BITUMINOUS STABILIZED SUBBASE TYPE I, II, OR III	TON			
52320	BITUMINOUS STABILIZED SUBBASE	TON			
52445	BITUMINOUS BASE	TON			
52446	BITUMINOUS BASE (SIZE NO 53B)	TON			
52451	BITUMINOUS BINDER	TON			
52450	BITUMINOUS SURFACE	TON			
52456	BITUMINOUS MATERIAL FOR TACK COAT	SYS.			680
52460	BITUMINOUS MATERIAL FOR PRIME COAT	TON			
52465	BITUMINOUS MATERIAL FOR SEAL COAT	TON			
52470	BITUMINOUS MIXTURE FOR APPROACHES	TON			552 545
52475	BITUMINOUS MIXTURE FOR SHOULDER	TON			
52480	BITUMINOUS MATERIAL, APPLIED	TON			
52500	GUARD RAIL, TYPE A	LFT.			
52505	GUARD RAIL, TYPE B	LFT.			
52510	GUARD RAIL, TYPE C	LFT.			
52515	GUARD RAIL, TYPE D	LFT.			
52520	GUARD RAIL, TYPE E	LFT.			
52525	GUARD RAIL, TYPE F	LFT.			
52530	GUARD RAIL, TYPE G	LFT.			
52538	GUARD RAIL, END TREATMENT, TYPE I	EACH			4
06035	RESET GUARD RAIL	LFT.			
52535	REMOVAL OF GUARD RAIL	LFT.			138
52380	SODDING	SYS.			191
52912	MULCHED SEEDING (LEGUME)	SYS.			2280
52388	SEED MIXTURE "R"	LBS.			
52397	SEED MIXTURE "RR"	LBS.			
1. 52400	MULCHING MATERIAL	TON			
2. 52305	FERTILIZER	TON			
52410	WATER	M.G.			
3. 52435	AGRICULTURAL LIMESTONE	TON			1
52398	SEED MIXTURE "CV"	LBS.			
52401	MULCHING MATERIAL (WOOD CELLULOSE FIBER)	TON			
	"DO NOT SPRAY" SIGN	EACH			4
52640	MAINTAINING TRAFFIC	LSUM			1
52370	CLEARING RIGHT-OF-WAY	LSUM			
52651	GUARD RAIL, TYPE H	LFT.			333 338

1. INCLUDES ____ TONS FOR SEED MIXTURE "R"
2. INCLUDES ____ TONS FOR SEED MIXTURE "R"
3. INCLUDES ____ TONS FOR SEED MIXTURE "R"

*Includes:
50 Tons for Wedge & Leveling
149 Tons for Shoulders
334 Tons for Bituminous Widening
Tons for Bituminous Drive

+ Includes 2 Tons for Conc. Pads.

APPROACH PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
07025	PIPE: GR. A (0.064" FBCCS) 12"	LFT.			
07075	PIPE: GR. A (0.064" FBCCS) 15"	LFT.			
07125	PIPE: GR. A (0.064" FBCCS) 18"	LFT.			
07175	PIPE: GR. A (0.064" FBCCS) 24"	LFT.			
07225	PIPE: GR. A (0.064" FBCCS) 30"	LFT.			
07275	PIPE: GR. A (0.064" FBCCS) 36"	LFT.			
07325	PIPE: GR. A (0.064" FBCCS) 42"	LFT.			
10000	PIPE: GR. D (0.064" CS) 12"	LFT.			15
10025	PIPE: GR. D (0.064" CS) 15"	LFT.			15
10050	PIPE: GR. D (0.064" CS) 18"	LFT.			185 193
10075	PIPE: GR. D (0.064" CS) 24"	LFT.			
10100	PIPE: GR. D (0.064" CS) 30"	LFT.			
10125	PIPE: GR. D (0.064" CS) 36"	LFT.			
10150	PIPE: GR. D (0.064" CS) 42"	LFT.			
	PIPE: GR. D (0.079" CS) 18"	LFT.			58
34000	PIPE: 0.052" FBC PERF. CS 6"	LFT.			
29000	PIPE: 0.064" FBCCS 12"	LFT.			
52375	CONCRETE CLASS A IN STRUCTURE	CYS.			
52376	CONCRETE CLASS C IN STRUCTURE	CYS.			
46000	PIPE END SECTION 12"	EACH			
46005	PIPE END SECTION 15"	EACH			
46010	PIPE END SECTION 18"	EACH			4
46015	PIPE END SECTION 21"	EACH			
46020	PIPE END SECTION 24"	EACH			
46025	PIPE END SECTION 27"	EACH			
46030	PIPE END SECTION 30"	EACH			
46035	PIPE END SECTION 33"	EACH			
46040	PIPE END SECTION 36"	EACH			
45000	INLET, TYPE A-1	EACH			
45025	INLET, TYPE D-6	EACH			
45030	INLET, TYPE E-7	EACH			
45070	INLET, TYPE P-12A	EACH			
06335	PAVED SIDE DITCH TYPE A	LFT.			
06340	PAVED SIDE DITCH TYPE B	LFT.			136
06345	PAVED SIDE DITCH TYPE C	LFT.			
06350	PAVED SIDE DITCH TYPE D	LFT.			
06355	PAVED SIDE DITCH TYPE E	LFT.			
06360	PAVED SIDE DITCH TYPE F	LFT.			
06365	PAVED SIDE DITCH TYPE G	LFT.			
	REMOVAL OF PAVED SIDE DITCH	LFT.			40
	W-BEAM GUARD RAIL TERMINAL SECTION	EACH			2

** INCLUDES ____ LFT. FOR YELLOW BARRIER LINE

APPROACH PAY ITEMS

CODE NO.	DESCRIPTION	UNIT	STRUCTURE		TOTAL QUANTITY
06040	42" FENCE, F. FIELD	LFT.			
06045	48" FENCE, CHAIN LINK	LFT.			
52340	CONSTRUCTION SIGNS (TYPE A)	EACH			18
52345	CONSTRUCTION SIGNS (TYPE B)	EACH			3
52350	STANDARD BARRICADES (TYPE III)	EACH			
06630	STOP SIGN, TYPE R-1A	EACH			
06652	DO NOT PASS SIGN, TYPE R-11-A	EACH			
06655	YIELD SIGN, TYPE R-301	EACH			
06657	PASS WITH CARE SIGN, TYPE R-12A	EACH			
06660	CURVE SIGN, TYPE W-2AR	EACH			
06665	CURVE SIGN, TYPE W-2AL	EACH			
06670	REVERSE CURVE SIGN, TYPE W-4AR	EACH			
06675	REVERSE CURVE SIGN, TYPE W-4AL	EACH			
06680	LARGE ARROW SIGN, TYPE W-11A	EACH			
06685	STOP AHEAD SIGN, TYPE W-13A	EACH			
06725	DELINEATOR WITH POST, TYPE D-1 3 INCH	EACH			
06740	DELINEATOR WITH POST, TYPE D-2 3 INCH	EACH			
06755	DELINEATOR WITH POST, TYPE D-3 3 INCH	EACH			
06770	DELINEATOR POST	EACH			
52366	TEMPORARY PAVEMENT MARKING TAPE	LFT.			
52367	TEMPORARY PAVEMENT MARKING PAINT	LFT.			
52360	RIGHT-OF-WAY MARKERS	EACH			
52365	PAINTED LINE **	LFT.			
06500	MONUMENT, TYPE A	EACH			
06505	MONUMENT, TYPE B	EACH			
06510	MONUMENT, TYPE C	EACH			
06515	MONUMENT, TYPE D	EACH			
52831	FLASHING ARROW SIGN	EACH			
53001	Line, Solid, White, 4"	LFT.			1440
53035	Line, Skip, Yellow, 4"	LFT.			200
53020	Removal of Line, Solid, White, 4"	LFT.			1155
53041	Removal of Line, Skip, Yellow, 4"	LFT.			90
52556	Temporary Concrete Barrier	LFT.			720
53072	Temporary Traffic Signals	LSUM			1
52361	Temporary Pavement Marking Type I	LFT.			3580
52362	Temporary Pavement Marking Type II	LFT.			2320

END STR

REVISIONS	
DATE	ITEM
4-17-85	Add: 10000, 10025
6-17-85	Pipe: Gr D (0.079" CS) 18", 52470, 52255, 52308, 52530
3-14-86	51002, 51030, 51032, 51833, 52300, 52308, Add: 53020, 52531 Deleted: 52530, 5M20
01-14-87	52308, 52470, 52531, 10050 W-Beam Guard Rail Terminal Section Added

SUMMARIZED MEB 9-11-84 C'K'D MHW 9-13-84

TRACED ____ C'K'D ____

MARCH, 1975

BRIDGE ESTIMATE OF QUANTITIES

INDIANA STATE HIGHWAY COMMISSION

FUNDING: FEDERAL 80% AND STATE 20%

DATE JANUARY 25, 1985

SUBMITTED FOR APPROVAL: John S. Spencer

PROJECT: FR-158-2(5)
CONTRACT NO: B-16317
BRIDGE FILE: 224-90-15454

SHEET 13 OF 43

JOHN S. SPENCER REGISTERED No. 9740 STATE OF INDIANA PROFESSIONAL ENGINEER